Pasangan Latin Square Komutatif Order 4

Terdapat 4752 pasangan (A, B) dengan $A \neq B$ yang memenuhi $A \otimes B = B \otimes A$:

1. $L_1 \circ L_{27}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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2. $L_1 \circ L_{30}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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3. $L_1 \circ L_{50}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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4. $L_1 \circ L_{53}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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5. $L_1 \circ L_{92}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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6. $L_1 \circ L_{94}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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7. $L_1 \circ L_{99}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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8. $L_1 \circ L_{102}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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9. $L_1 \circ L_{104}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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10. $L_1 \circ L_{131}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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11. $L_1 \circ L_{133}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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12. $L_1 \circ L_{148}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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13. $L_1 \circ L_{159}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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14. $L_1 \circ L_{181}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \\ 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

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15. $L_1 \circ L_{193}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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16. $L_1 \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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17. $L_1 \circ L_{216}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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18. $L_1 \circ L_{235}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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19. $L_1 \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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20. $L_1 \circ L_{259}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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21. $L_1 \circ L_{261}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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22. $L_1 \circ L_{273}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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23. $L_1 \circ L_{276}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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24. $L_1 \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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25. $L_1 \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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26. $L_1 \circ L_{323}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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27. $L_1 \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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28. $L_1 \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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29. $L_1 \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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30. $L_1 \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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31. $L_1 \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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32. $L_1 \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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33. $L_1 \circ L_{405}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

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34. $L_1 \circ L_{420}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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35. $L_1 \circ L_{423}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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36. $L_1 \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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37. $L_1 \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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38. $L_1 \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

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39. $L_1 \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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40. $L_1 \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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41. $L_1 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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42. $L_1 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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43. $L_1 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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44. $L_1 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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45. $L_1 \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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46. $L_1 \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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47. $L_1 \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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48. $L_2 \circ L_{50}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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49. $L_2 \circ L_{53}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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50. $L_2 \circ L_{92}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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51. $L_2 \circ L_{94}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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52. $L_2 \circ L_{99}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

53. $L_2 \circ L_{102}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

54. $L_2 \circ L_{104}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

55. $L_2 \circ L_{131}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

56. $L_2 \circ L_{133}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

---

57. $L_2 \circ L_{182}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

---

58. $L_2 \circ L_{193}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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59. $L_2 \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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60. $L_2 \circ L_{235}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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61. $L_2 \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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62. $L_2 \circ L_{259}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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63. $L_2 \circ L_{261}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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64. $L_2 \circ L_{273}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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65. $L_2 \circ L_{276}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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66. $L_2 \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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67. $L_2 \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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68. $L_2 \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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69. $L_2 \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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70. $L_2 \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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71. $L_2 \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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72. $L_2 \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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73. $L_2 \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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74. $L_2 \circ L_{405}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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75. $L_2 \circ L_{420}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

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76. $L_2 \circ L_{423}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

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77. $L_2 \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

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78. $L_2 \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

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79. $L_2 \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

---

80. $L_2 \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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81. $L_2 \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

82. $L_2 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

83. $L_2 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

84. $L_2 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

85. $L_2 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

---

86. $L_2 \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

87. $L_2 \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

88. $L_2 \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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89. $L_3 \circ L_{25}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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90. $L_3 \circ L_{52}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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91. $L_3 \circ L_{90}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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92. $L_3 \circ L_{97}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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93. $L_3 \circ L_{135}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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94. $L_3 \circ L_{146}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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95. $L_3 \circ L_{183}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \\ 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \end{pmatrix}$$

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96. $L_3 \circ L_{197}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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97. $L_3 \circ L_{239}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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98. $L_3 \circ L_{263}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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99. $L_3 \circ L_{278}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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100. $L_3 \circ L_{290}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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101. $L_3 \circ L_{327}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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102. $L_3 \circ L_{341}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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103. $L_3 \circ L_{383}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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104. $L_3 \circ L_{407}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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105. $L_3 \circ L_{422}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

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106. $L_3 \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

---

107. $L_3 \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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108. $L_3 \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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109. $L_3 \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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110. $L_3 \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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111. $L_3 \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

112. $L_4 \circ L_{52}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

---

113. $L_4 \circ L_{90}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

---

114. $L_4 \circ L_{97}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

---

115. $L_4 \circ L_{135}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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116. $L_4 \circ L_{184}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

---

117. $L_4 \circ L_{197}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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118. $L_4 \circ L_{239}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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119. $L_4 \circ L_{263}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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120. $L_4 \circ L_{278}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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121. $L_4 \circ L_{290}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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122. $L_4 \circ L_{327}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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123. $L_4 \circ L_{341}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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124. $L_4 \circ L_{383}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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125. $L_4 \circ L_{407}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

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126. $L_4 \circ L_{422}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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127. $L_4 \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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128. $L_4 \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

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129. $L_4 \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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130. $L_4 \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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131. $L_4 \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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132. $L_4 \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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133. $L_5 \circ L_{27}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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134. $L_5 \circ L_{30}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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135. $L_5 \circ L_{92}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

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136. $L_5 \circ L_{94}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

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137. $L_5 \circ L_{150}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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138. $L_5 \circ L_{181}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 7 \\ 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

---

139. $L_5 \circ L_{235}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

---

140. $L_5 \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

---

141. $L_5 \circ L_{323}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

142. $L_5 \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

143. $L_5 \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

144. $L_5 \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

145. $L_5 \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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146. $L_5 \circ L_{401}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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147. $L_5 \circ L_{417}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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148. $L_5 \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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149. $L_5 \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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150. $L_5 \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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151. $L_5 \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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152. $L_5 \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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153. $L_5 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 5 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

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154. $L_5 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

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155. $L_5 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 5 \\ 7 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

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156. $L_5 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 7 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

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157. $L_5 \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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158. $L_5 \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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159. $L_5 \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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160. $L_6 \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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161. $L_6 \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

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162. $L_6 \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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163. $L_6 \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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164. $L_7 \circ L_{152}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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165. $L_7 \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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166. $L_7 \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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167. $L_7 \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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168. $L_7 \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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169. $L_7 \circ L_{439}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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170. $L_7 \circ L_{467}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 7 & 8 \\ 7 & 5 & 8 & 7 \\ 7 & 8 & 5 & 7 \\ 8 & 7 & 7 & 5 \end{pmatrix}$$

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171. $L_7 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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172. $L_7 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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173. $L_7 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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174. $L_7 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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175. $L_9 \circ L_{42}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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176. $L_9 \circ L_{45}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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177. $L_9 \circ L_{57}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 5 \\ 7 & 8 & 5 & 7 \\ 7 & 5 & 8 & 7 \\ 5 & 7 & 7 & 8 \end{pmatrix}$$

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178. $L_9 \circ L_{74}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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179. $L_9 \circ L_{77}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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180. $L_9 \circ L_{105}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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181. $L_9 \circ L_{107}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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182. $L_9 \circ L_{123}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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183. $L_9 \circ L_{128}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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184. $L_9 \circ L_{148}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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185. $L_9 \circ L_{159}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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186. $L_9 \circ L_{176}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 7 \\ 5 & 7 & 7 & 8 \\ 8 & 7 & 7 & 5 \\ 7 & 5 & 8 & 7 \end{pmatrix}$$

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187. $L_9 \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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188. $L_9 \circ L_{216}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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189. $L_9 \circ L_{227}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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190. $L_9 \circ L_{232}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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191. $L_9 \circ L_{254}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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192. $L_9 \circ L_{256}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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193. $L_9 \circ L_{265}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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194. $L_9 \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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195. $L_9 \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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196. $L_9 \circ L_{315}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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197. $L_9 \circ L_{320}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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198. $L_9 \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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199. $L_9 \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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200. $L_9 \circ L_{376}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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201. $L_9 \circ L_{400}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 7 \\ 8 & 7 & 7 & 5 \\ 5 & 7 & 7 & 8 \\ 7 & 8 & 5 & 7 \end{pmatrix}$$

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202. $L_9 \circ L_{409}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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203. $L_9 \circ L_{411}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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204. $L_9 \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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205. $L_9 \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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206. $L_9 \circ L_{464}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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207. $L_9 \circ L_{503}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 7 & 8 \\ 7 & 5 & 8 & 7 \\ 7 & 8 & 5 & 7 \\ 8 & 7 & 7 & 5 \end{pmatrix}$$

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208. $L_9 \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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209. $L_9 \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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210. $L_9 \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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211. $L_9 \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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212. $L_9 \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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213. $L_9 \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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214. $L_{10} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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215. $L_{10} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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216. $L_{11} \circ L_{15}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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217. $L_{11} \circ L_{153}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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218. $L_{11} \circ L_{156}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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219. $L_{11} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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220. $L_{11} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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221. $L_{11} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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222. $L_{11} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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223. $L_{11} \circ L_{444}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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224. $L_{11} \circ L_{450}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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225. $L_{11} \circ L_{459}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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226. $L_{11} \circ L_{479}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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227. $L_{11} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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228. $L_{11} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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229. $L_{11} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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230. $L_{11} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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231. $L_{14} \circ L_{16}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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232. $L_{14} \circ L_{154}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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233. $L_{14} \circ L_{157}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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234. $L_{15} \circ L_{153}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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235. $L_{15} \circ L_{156}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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236. $L_{15} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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237. $L_{15} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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238. $L_{15} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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239. $L_{15} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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240. $L_{15} \circ L_{444}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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241. $L_{15} \circ L_{450}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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242. $L_{15} \circ L_{459}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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243. $L_{15} \circ L_{479}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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244. $L_{15} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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245. $L_{15} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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246. $L_{15} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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247. $L_{15} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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248. $L_{16} \circ L_{154}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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249. $L_{16} \circ L_{157}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (132)(4) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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250. $L_{18} \circ L_{34}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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251. $L_{18} \circ L_{65}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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252. $L_{18} \circ L_{70}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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253. $L_{18} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

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254. $L_{18} \circ L_{115}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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255. $L_{18} \circ L_{119}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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256. $L_{18} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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257. $L_{18} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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258. $L_{18} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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259. $L_{18} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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260. $L_{18} \circ L_{206}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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261. $L_{18} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 5 \\ 5 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

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262. $L_{18} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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263. $L_{18} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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264. $L_{18} \circ L_{286}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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265. $L_{18} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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266. $L_{18} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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267. $L_{18} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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268. $L_{18} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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269. $L_{18} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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270. $L_{18} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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271. $L_{18} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 5 \\ 5 & 6 & 7 & 8 \end{pmatrix}$$

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272. $L_{18} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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273. $L_{18} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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274. $L_{18} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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275. $L_{18} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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276. $L_{18} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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277. $L_{18} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

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278. $L_{18} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

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279. $L_{18} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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280. $L_{18} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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281. $L_{19} \circ L_{21}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (134)(2) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (134)(2) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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282. $L_{19} \circ L_{162}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (134)(2) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (134)(2) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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283. $L_{19} \circ L_{166}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1243) \\ \tau_3 &= (134)(2) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (134)(2) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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284. $L_{20} \circ L_{24}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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285. $L_{20} \circ L_{161}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

286. $L_{20} \circ L_{163}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

287. $L_{20} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

288. $L_{20} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

289. $L_{20} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

290. $L_{20} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

291. $L_{20} \circ L_{386}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

292. $L_{20} \circ L_{393}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

293. $L_{20} \circ L_{416}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

294. $L_{20} \circ L_{430}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

295. $L_{20} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

296. $L_{20} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

297. $L_{20} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

298. $L_{20} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

299. $L_{20} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

300. $L_{20} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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301. $L_{21} \circ L_{162}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (142)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (142)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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302. $L_{21} \circ L_{166}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (142)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (134)(2) \\ \tau_4 = (142)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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303. $L_{22} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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304. $L_{22} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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305. $L_{24} \circ L_{34}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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306. $L_{24} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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307. $L_{24} \circ L_{161}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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308. $L_{24} \circ L_{163}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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309. $L_{24} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

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310. $L_{24} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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311. $L_{24} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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312. $L_{24} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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313. $L_{24} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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314. $L_{24} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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315. $L_{24} \circ L_{386}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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316. $L_{24} \circ L_{393}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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317. $L_{24} \circ L_{416}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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318. $L_{24} \circ L_{430}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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319. $L_{24} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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320. $L_{24} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

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321. $L_{24} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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322. $L_{24} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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323. $L_{24} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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324. $L_{24} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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325. $L_{24} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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326. $L_{24} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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327. $L_{25} \circ L_{52}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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328. $L_{25} \circ L_{90}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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329. $L_{25} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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330. $L_{25} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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331. $L_{25} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \\ 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \end{pmatrix}$$

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332. $L_{25} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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333. $L_{25} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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334. $L_{25} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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335. $L_{25} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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336. $L_{25} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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337. $L_{25} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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338. $L_{25} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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339. $L_{25} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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340. $L_{25} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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341. $L_{25} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

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342. $L_{25} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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343. $L_{25} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

---

344. $L_{25} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

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345. $L_{25} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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346. $L_{25} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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347. $L_{25} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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348. $L_{25} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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349. $L_{26} \circ L_{52}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

---

350. $L_{26} \circ L_{90}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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351. $L_{26} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

---

352. $L_{26} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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353. $L_{26} \circ L_{145}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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354. $L_{26} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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355. $L_{26} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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356. $L_{26} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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357. $L_{26} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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358. $L_{26} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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359. $L_{26} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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360. $L_{26} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

361. $L_{26} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

362. $L_{26} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

363. $L_{26} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

364. $L_{26} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

365. $L_{26} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

366. $L_{26} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

367. $L_{26} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

368. $L_{26} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

369. $L_{26} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

370. $L_{27} \circ L_{50}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

371. $L_{27} \circ L_{53}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

372. $L_{27} \circ L_{92}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

373. $L_{27} \circ L_{94}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

374. $L_{27} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

375. $L_{27} \circ L_{102}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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376. $L_{27} \circ L_{129}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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377. $L_{27} \circ L_{131}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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378. $L_{27} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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379. $L_{27} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \\ 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

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380. $L_{27} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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381. $L_{27} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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382. $L_{27} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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383. $L_{27} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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384. $L_{27} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

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385. $L_{27} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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386. $L_{27} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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387. $L_{27} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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388. $L_{27} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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389. $L_{27} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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390. $L_{27} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

391. $L_{27} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

392. $L_{27} \circ L_{294}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

393. $L_{27} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

394. $L_{27} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

395. $L_{27} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

396. $L_{27} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

397. $L_{27} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

398. $L_{27} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

399. $L_{27} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

400. $L_{27} \circ L_{401}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

401. $L_{27} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

402. $L_{27} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

403. $L_{27} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

404. $L_{27} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

405. $L_{27} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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406. $L_{27} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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407. $L_{27} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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408. $L_{27} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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409. $L_{27} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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410. $L_{27} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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411. $L_{27} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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412. $L_{27} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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413. $L_{27} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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414. $L_{27} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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415. $L_{28} \circ L_{50}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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416. $L_{28} \circ L_{53}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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417. $L_{28} \circ L_{92}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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418. $L_{28} \circ L_{94}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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419. $L_{28} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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420. $L_{28} \circ L_{102}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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421. $L_{28} \circ L_{129}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

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422. $L_{28} \circ L_{131}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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423. $L_{28} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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424. $L_{28} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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425. $L_{28} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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426. $L_{28} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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427. $L_{28} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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428. $L_{28} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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429. $L_{28} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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430. $L_{28} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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431. $L_{28} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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432. $L_{28} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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433. $L_{28} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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434. $L_{28} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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435. $L_{28} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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436. $L_{28} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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437. $L_{28} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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438. $L_{28} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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439. $L_{28} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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440. $L_{28} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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441. $L_{28} \circ L_{401}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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442. $L_{28} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

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443. $L_{28} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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444. $L_{28} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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445. $L_{28} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

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446. $L_{28} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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447. $L_{28} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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448. $L_{28} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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449. $L_{28} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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450. $L_{28} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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451. $L_{28} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

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452. $L_{28} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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453. $L_{28} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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454. $L_{28} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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455. $L_{28} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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456. $L_{29} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (143)(2) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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457. $L_{29} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (143)(2) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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458. $L_{29} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (143)(2) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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459. $L_{29} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (143)(2) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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460. $L_{30} \circ L_{50}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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461. $L_{30} \circ L_{53}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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462. $L_{30} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

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463. $L_{30} \circ L_{179}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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464. $L_{30} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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465. $L_{30} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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466. $L_{30} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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467. $L_{30} \circ L_{294}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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468. $L_{30} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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469. $L_{30} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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470. $L_{30} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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471. $L_{30} \circ L_{403}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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472. $L_{30} \circ L_{423}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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473. $L_{30} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

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474. $L_{30} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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475. $L_{30} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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476. $L_{30} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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477. $L_{30} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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478. $L_{30} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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479. $L_{30} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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480. $L_{30} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

481. $L_{30} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

482. $L_{30} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

483. $L_{30} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 5 & 8 \\ 8 & 5 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

484. $L_{30} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

485. $L_{32} \circ L_{177}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

486. $L_{32} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

487. $L_{32} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

488. $L_{32} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

489. $L_{32} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

490. $L_{32} \circ L_{437}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \end{pmatrix}$$

491. $L_{32} \circ L_{465}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

492. $L_{32} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

493. $L_{32} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

494. $L_{32} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

495. $L_{32} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

496. $L_{33} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (123)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

497. $L_{33} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (123)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

498. $L_{34} \circ L_{65}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

499. $L_{34} \circ L_{70}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

500. $L_{34} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \end{pmatrix}$$

501. $L_{34} \circ L_{115}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

502. $L_{34} \circ L_{119}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

503. $L_{34} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

504. $L_{34} \circ L_{142}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

505. $L_{34} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \end{pmatrix}$$

506. $L_{34} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

507. $L_{34} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

508. $L_{34} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

509. $L_{34} \circ L_{206}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

510. $L_{34} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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511. $L_{34} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

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512. $L_{34} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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513. $L_{34} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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514. $L_{34} \circ L_{286}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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515. $L_{34} \circ L_{307}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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516. $L_{34} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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517. $L_{34} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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518. $L_{34} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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519. $L_{34} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

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520. $L_{34} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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521. $L_{34} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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522. $L_{34} \circ L_{386}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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523. $L_{34} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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524. $L_{34} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \end{pmatrix}$$

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525. $L_{34} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

526. $L_{34} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

527. $L_{34} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

528. $L_{34} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

529. $L_{34} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

530. $L_{34} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \end{pmatrix}$$

531. $L_{34} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

532. $L_{34} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

533. $L_{34} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

534. $L_{34} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

535. $L_{35} \circ L_{39}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

536. $L_{35} \circ L_{187}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

537. $L_{35} \circ L_{191}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

538. $L_{38} \circ L_{40}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

539. $L_{38} \circ L_{190}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

540. $L_{38} \circ L_{192}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

541. $L_{38} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

542. $L_{38} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

543. $L_{38} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

544. $L_{38} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

545. $L_{38} \circ L_{442}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

546. $L_{38} \circ L_{451}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

547. $L_{38} \circ L_{461}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

548. $L_{38} \circ L_{478}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

549. $L_{38} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

550. $L_{38} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

551. $L_{38} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

552. $L_{38} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

553. $L_{39} \circ L_{187}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

554. $L_{39} \circ L_{191}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

555. $L_{40} \circ L_{190}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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556. $L_{40} \circ L_{192}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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557. $L_{40} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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558. $L_{40} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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559. $L_{40} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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560. $L_{40} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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561. $L_{40} \circ L_{442}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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562. $L_{40} \circ L_{451}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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563. $L_{40} \circ L_{461}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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564. $L_{40} \circ L_{478}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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565. $L_{40} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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566. $L_{40} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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567. $L_{40} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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568. $L_{40} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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569. $L_{42} \circ L_{57}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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570. $L_{42} \circ L_{74}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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571. $L_{42} \circ L_{77}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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572. $L_{42} \circ L_{105}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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573. $L_{42} \circ L_{123}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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574. $L_{42} \circ L_{128}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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575. $L_{42} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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576. $L_{42} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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577. $L_{42} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 6 & 7 & 5 & 8 \\ 8 & 5 & 7 & 6 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

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578. $L_{42} \circ L_{227}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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579. $L_{42} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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580. $L_{42} \circ L_{254}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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581. $L_{42} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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582. $L_{42} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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583. $L_{42} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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584. $L_{42} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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585. $L_{42} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

586. $L_{42} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

587. $L_{42} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

588. $L_{42} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

589. $L_{42} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

590. $L_{42} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 6 \\ 7 & 8 & 6 & 5 \\ 5 & 6 & 8 & 7 \\ 6 & 7 & 5 & 8 \end{pmatrix}$$

591. $L_{42} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

592. $L_{42} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

593. $L_{42} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

594. $L_{42} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

595. $L_{42} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

596. $L_{42} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

597. $L_{42} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

598. $L_{42} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

599. $L_{43} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

600. $L_{43} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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601. $L_{44} \circ L_{47}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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602. $L_{44} \circ L_{170}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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603. $L_{44} \circ L_{173}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (124)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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604. $L_{45} \circ L_{48}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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605. $L_{45} \circ L_{57}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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606. $L_{45} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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607. $L_{45} \circ L_{171}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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608. $L_{45} \circ L_{174}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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609. $L_{45} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

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610. $L_{45} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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611. $L_{45} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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612. $L_{45} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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613. $L_{45} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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614. $L_{45} \circ L_{390}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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615. $L_{45} \circ L_{398}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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616. $L_{45} \circ L_{411}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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617. $L_{45} \circ L_{431}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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618. $L_{45} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 5 & 6 \\ 6 & 5 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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619. $L_{45} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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620. $L_{45} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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621. $L_{45} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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622. $L_{45} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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623. $L_{45} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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624. $L_{45} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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625. $L_{45} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

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626. $L_{47} \circ L_{170}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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627. $L_{47} \circ L_{173}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1234) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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628. $L_{48} \circ L_{171}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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629. $L_{48} \circ L_{174}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (123)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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630. $L_{48} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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631. $L_{48} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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632. $L_{48} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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633. $L_{48} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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634. $L_{48} \circ L_{390}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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635. $L_{48} \circ L_{398}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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636. $L_{48} \circ L_{411}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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637. $L_{48} \circ L_{431}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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638. $L_{48} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 5 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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639. $L_{48} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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640. $L_{48} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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641. $L_{48} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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642. $L_{48} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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643. $L_{48} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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644. $L_{49} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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645. $L_{49} \circ L_{104}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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646. $L_{49} \circ L_{131}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

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647. $L_{49} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

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648. $L_{49} \circ L_{196}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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649. $L_{49} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

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650. $L_{49} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

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651. $L_{49} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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652. $L_{49} \circ L_{326}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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653. $L_{49} \circ L_{382}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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654. $L_{49} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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655. $L_{49} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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656. $L_{49} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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657. $L_{49} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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658. $L_{49} \circ L_{423}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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659. $L_{49} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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660. $L_{49} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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661. $L_{49} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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662. $L_{49} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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663. $L_{49} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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664. $L_{49} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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665. $L_{49} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

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666. $L_{49} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

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667. $L_{49} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

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668. $L_{49} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

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669. $L_{49} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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670. $L_{49} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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671. $L_{50} \circ L_{92}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

---

672. $L_{50} \circ L_{94}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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673. $L_{50} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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674. $L_{50} \circ L_{104}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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675. $L_{50} \circ L_{131}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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676. $L_{50} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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677. $L_{50} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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678. $L_{50} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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679. $L_{50} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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680. $L_{50} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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681. $L_{50} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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682. $L_{50} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

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683. $L_{50} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

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684. $L_{50} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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685. $L_{50} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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686. $L_{50} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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687. $L_{50} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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688. $L_{50} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

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689. $L_{50} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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690. $L_{50} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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691. $L_{50} \circ L_{323}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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692. $L_{50} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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693. $L_{50} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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694. $L_{50} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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695. $L_{50} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \\ 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

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696. $L_{50} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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697. $L_{50} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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698. $L_{50} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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699. $L_{50} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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700. $L_{50} \circ L_{423}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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701. $L_{50} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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702. $L_{50} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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703. $L_{50} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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704. $L_{50} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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705. $L_{50} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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706. $L_{50} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

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707. $L_{50} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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708. $L_{50} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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709. $L_{50} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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710. $L_{50} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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711. $L_{50} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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712. $L_{50} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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713. $L_{51} \circ L_{54}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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714. $L_{51} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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715. $L_{51} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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716. $L_{51} \circ L_{194}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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717. $L_{51} \circ L_{198}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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718. $L_{51} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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719. $L_{51} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

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720. $L_{51} \circ L_{328}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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721. $L_{51} \circ L_{329}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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722. $L_{51} \circ L_{368}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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723. $L_{51} \circ L_{384}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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724. $L_{51} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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725. $L_{51} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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726. $L_{51} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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727. $L_{51} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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728. $L_{51} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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729. $L_{51} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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730. $L_{51} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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731. $L_{51} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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732. $L_{51} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

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733. $L_{51} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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734. $L_{51} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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735. $L_{51} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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736. $L_{52} \circ L_{90}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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737. $L_{52} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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738. $L_{52} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

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739. $L_{52} \circ L_{145}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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740. $L_{52} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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741. $L_{52} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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742. $L_{52} \circ L_{184}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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743. $L_{52} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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744. $L_{52} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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745. $L_{52} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

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746. $L_{52} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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747. $L_{52} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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748. $L_{52} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

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749. $L_{52} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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750. $L_{52} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \\ 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \end{pmatrix}$$

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751. $L_{52} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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752. $L_{52} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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753. $L_{52} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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754. $L_{52} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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755. $L_{52} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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756. $L_{52} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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757. $L_{52} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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758. $L_{52} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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759. $L_{52} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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760. $L_{52} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

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761. $L_{52} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

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762. $L_{52} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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763. $L_{53} \circ L_{92}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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764. $L_{53} \circ L_{94}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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765. $L_{53} \circ L_{131}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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766. $L_{53} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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767. $L_{53} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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768. $L_{53} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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769. $L_{53} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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770. $L_{53} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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771. $L_{53} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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772. $L_{53} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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773. $L_{53} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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774. $L_{53} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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775. $L_{53} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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776. $L_{53} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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777. $L_{53} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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778. $L_{53} \circ L_{323}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

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779. $L_{53} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

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780. $L_{53} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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781. $L_{53} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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782. $L_{53} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 6 \\ 5 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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783. $L_{53} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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784. $L_{53} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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785. $L_{53} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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786. $L_{53} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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787. $L_{53} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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788. $L_{53} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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789. $L_{53} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 5 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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790. $L_{53} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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791. $L_{53} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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792. $L_{53} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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793. $L_{53} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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794. $L_{53} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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795. $L_{53} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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796. $L_{53} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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797. $L_{53} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

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798. $L_{53} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 5 \\ 6 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

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799. $L_{54} \circ L_{194}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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800. $L_{54} \circ L_{198}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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801. $L_{54} \circ L_{328}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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802. $L_{54} \circ L_{329}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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803. $L_{54} \circ L_{368}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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804. $L_{54} \circ L_{384}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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805. $L_{54} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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806. $L_{54} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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807. $L_{54} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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808. $L_{54} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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809. $L_{54} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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810. $L_{54} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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811. $L_{54} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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812. $L_{54} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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813. $L_{54} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

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814. $L_{54} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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815. $L_{55} \circ L_{199}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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816. $L_{55} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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817. $L_{55} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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818. $L_{55} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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819. $L_{55} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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820. $L_{55} \circ L_{490}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 7 & 8 \\ 7 & 5 & 8 & 7 \\ 7 & 8 & 5 & 7 \\ 8 & 7 & 7 & 5 \end{pmatrix}$$

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821. $L_{55} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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822. $L_{55} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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823. $L_{55} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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824. $L_{55} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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825. $L_{55} \circ L_{563}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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826. $L_{57} \circ L_{74}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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827. $L_{57} \circ L_{77}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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828. $L_{57} \circ L_{105}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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829. $L_{57} \circ L_{107}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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830. $L_{57} \circ L_{123}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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831. $L_{57} \circ L_{128}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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832. $L_{57} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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833. $L_{57} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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834. $L_{57} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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835. $L_{57} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

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836. $L_{57} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

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837. $L_{57} \circ L_{227}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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838. $L_{57} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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839. $L_{57} \circ L_{254}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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840. $L_{57} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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841. $L_{57} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 7 \\ 8 & 7 & 7 & 5 \\ 5 & 7 & 7 & 8 \\ 7 & 8 & 5 & 7 \end{pmatrix}$$

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842. $L_{57} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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843. $L_{57} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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844. $L_{57} \circ L_{315}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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845. $L_{57} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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846. $L_{57} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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847. $L_{57} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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848. $L_{57} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 7 \\ 5 & 7 & 7 & 8 \\ 8 & 7 & 7 & 5 \\ 7 & 5 & 8 & 7 \end{pmatrix}$$

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849. $L_{57} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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850. $L_{57} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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851. $L_{57} \circ L_{411}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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852. $L_{57} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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853. $L_{57} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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854. $L_{57} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 7 & 8 \\ 7 & 5 & 8 & 7 \\ 7 & 8 & 5 & 7 \\ 8 & 7 & 7 & 5 \end{pmatrix}$$

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855. $L_{57} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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856. $L_{57} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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857. $L_{57} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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858. $L_{57} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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859. $L_{57} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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860. $L_{57} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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861. $L_{57} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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862. $L_{58} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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863. $L_{58} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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864. $L_{59} \circ L_{60}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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865. $L_{59} \circ L_{213}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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866. $L_{59} \circ L_{214}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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867. $L_{59} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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868. $L_{59} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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869. $L_{59} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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870. $L_{59} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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871. $L_{59} \circ L_{499}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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872. $L_{59} \circ L_{501}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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873. $L_{59} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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874. $L_{59} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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875. $L_{59} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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876. $L_{59} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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877. $L_{59} \circ L_{553}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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878. $L_{59} \circ L_{555}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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879. $L_{60} \circ L_{213}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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880. $L_{60} \circ L_{214}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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881. $L_{60} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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882. $L_{60} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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883. $L_{60} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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884. $L_{60} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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885. $L_{60} \circ L_{499}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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886. $L_{60} \circ L_{501}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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887. $L_{60} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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888. $L_{60} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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889. $L_{60} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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890. $L_{60} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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891. $L_{60} \circ L_{553}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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892. $L_{60} \circ L_{555}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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893. $L_{61} \circ L_{62}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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894. $L_{61} \circ L_{211}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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895. $L_{61} \circ L_{212}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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896. $L_{62} \circ L_{211}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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897. $L_{62} \circ L_{212}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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898. $L_{65} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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899. $L_{65} \circ L_{119}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

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900. $L_{65} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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901. $L_{65} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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902. $L_{65} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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903. $L_{65} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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904. $L_{65} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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905. $L_{65} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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906. $L_{65} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 5 \\ 5 & 7 & 6 & 8 \end{pmatrix}$$

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907. $L_{65} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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908. $L_{65} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 5 \\ 5 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

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909. $L_{65} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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910. $L_{65} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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911. $L_{65} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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912. $L_{65} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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913. $L_{65} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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914. $L_{65} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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915. $L_{65} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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916. $L_{65} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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917. $L_{65} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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918. $L_{65} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

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919. $L_{66} \circ L_{71}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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920. $L_{66} \circ L_{201}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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921. $L_{66} \circ L_{203}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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922. $L_{67} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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923. $L_{67} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

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924. $L_{67} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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925. $L_{67} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

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926. $L_{70} \circ L_{81}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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927. $L_{70} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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928. $L_{70} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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929. $L_{70} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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930. $L_{70} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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931. $L_{70} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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932. $L_{70} \circ L_{286}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

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933. $L_{70} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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934. $L_{70} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

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935. $L_{70} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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936. $L_{70} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

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937. $L_{70} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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938. $L_{70} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

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939. $L_{70} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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940. $L_{70} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

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941. $L_{70} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

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942. $L_{70} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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943. $L_{70} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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944. $L_{70} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

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945. $L_{71} \circ L_{201}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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946. $L_{71} \circ L_{203}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

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947. $L_{72} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

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948. $L_{72} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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949. $L_{73} \circ L_{76}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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950. $L_{73} \circ L_{226}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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951. $L_{73} \circ L_{231}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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952. $L_{74} \circ L_{105}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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953. $L_{74} \circ L_{128}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

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954. $L_{74} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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955. $L_{74} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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956. $L_{74} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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957. $L_{74} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

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958. $L_{74} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 6 & 7 \\ 6 & 8 & 7 & 5 \\ 5 & 7 & 8 & 6 \\ 7 & 6 & 5 & 8 \end{pmatrix}$$

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959. $L_{74} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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960. $L_{74} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 7 & 6 & 5 & 8 \\ 8 & 5 & 6 & 7 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

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961. $L_{74} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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962. $L_{74} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

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963. $L_{74} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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964. $L_{74} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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965. $L_{74} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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966. $L_{74} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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967. $L_{74} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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968. $L_{74} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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969. $L_{74} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

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970. $L_{74} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

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971. $L_{74} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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972. $L_{75} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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973. $L_{75} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

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974. $L_{76} \circ L_{226}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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975. $L_{76} \circ L_{231}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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976. $L_{77} \circ L_{105}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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977. $L_{77} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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978. $L_{77} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

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979. $L_{77} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

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980. $L_{77} \circ L_{254}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

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981. $L_{77} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

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982. $L_{77} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

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983. $L_{77} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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984. $L_{77} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

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985. $L_{77} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

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986. $L_{77} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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987. $L_{77} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

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988. $L_{77} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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989. $L_{77} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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990. $L_{77} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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991. $L_{77} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 6 & 5 & 7 \\ 7 & 5 & 6 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

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992. $L_{77} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

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993. $L_{77} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

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994. $L_{80} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

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995. $L_{80} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 5 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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996. $L_{80} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

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997. $L_{80} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

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998. $L_{81} \circ L_{115}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

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999. $L_{81} \circ L_{119}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1000. $L_{81} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1001. $L_{81} \circ L_{142}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1002. $L_{81} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1003. $L_{81} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

1004. $L_{81} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1005. $L_{81} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1006. $L_{81} \circ L_{206}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1007. $L_{81} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

1008. $L_{81} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1009. $L_{81} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \end{pmatrix}$$

1010. $L_{81} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1011. $L_{81} \circ L_{286}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1012. $L_{81} \circ L_{307}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1013. $L_{81} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1014. $L_{81} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1015. $L_{81} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1016. $L_{81} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \end{pmatrix}$$

1017. $L_{81} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1018. $L_{81} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1019. $L_{81} \circ L_{386}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1020. $L_{81} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1021. $L_{81} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1022. $L_{81} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \end{pmatrix}$$

1023. $L_{81} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1024. $L_{81} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1025. $L_{81} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1026. $L_{81} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1027. $L_{81} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1028. $L_{81} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1029. $L_{81} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1030. $L_{81} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1031. $L_{81} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1032. $L_{82} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1033. $L_{82} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1034. $L_{85} \circ L_{86}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1035. $L_{85} \circ L_{219}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1036. $L_{85} \circ L_{220}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1037. $L_{85} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1038. $L_{85} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1039. $L_{85} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1040. $L_{85} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1041. $L_{85} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1042. $L_{85} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1043. $L_{85} \circ L_{524}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1044. $L_{85} \circ L_{526}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1045. $L_{85} \circ L_{530}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1046. $L_{85} \circ L_{532}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1047. $L_{85} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1048. $L_{85} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1049. $L_{86} \circ L_{219}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1050. $L_{86} \circ L_{220}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1051. $L_{86} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1052. $L_{86} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1053. $L_{86} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1054. $L_{86} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1055. $L_{86} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1056. $L_{86} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1057. $L_{86} \circ L_{524}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1058. $L_{86} \circ L_{526}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1059. $L_{86} \circ L_{530}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1060. $L_{86} \circ L_{532}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1061. $L_{86} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1062. $L_{86} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1063. $L_{87} \circ L_{88}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1064. $L_{87} \circ L_{221}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1065. $L_{87} \circ L_{222}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (123)(4) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1066. $L_{88} \circ L_{221}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (123)(4) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1067. $L_{88} \circ L_{222}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (123)(4) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (123)(4) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1068. $L_{89} \circ L_{93}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1069. $L_{89} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1070. $L_{89} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1071. $L_{89} \circ L_{236}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1072. $L_{89} \circ L_{240}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1073. $L_{89} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

1074. $L_{89} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1075. $L_{89} \circ L_{289}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1076. $L_{89} \circ L_{304}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1077. $L_{89} \circ L_{342}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1078. $L_{89} \circ L_{359}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1079. $L_{89} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1080. $L_{89} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1081. $L_{89} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1082. $L_{89} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1083. $L_{89} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1084. $L_{89} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1085. $L_{89} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1086. $L_{89} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1087. $L_{89} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1088. $L_{89} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

1089. $L_{89} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1090. $L_{89} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1091. $L_{90} \circ L_{97}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1092. $L_{90} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1093. $L_{90} \circ L_{145}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1094. $L_{90} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1095. $L_{90} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1096. $L_{90} \circ L_{184}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1097. $L_{90} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1098. $L_{90} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1099. $L_{90} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1100. $L_{90} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

1101. $L_{90} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

1102. $L_{90} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1103. $L_{90} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \\ 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \end{pmatrix}$$

1104. $L_{90} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1105. $L_{90} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1106. $L_{90} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1107. $L_{90} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1108. $L_{90} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1109. $L_{90} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1110. $L_{90} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1111. $L_{90} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1112. $L_{90} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1113. $L_{90} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

1114. $L_{90} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1115. $L_{90} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1116. $L_{90} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1117. $L_{91} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1118. $L_{91} \circ L_{102}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

1119. $L_{91} \circ L_{129}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1120. $L_{91} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1121. $L_{91} \circ L_{238}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1122. $L_{91} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1123. $L_{91} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1124. $L_{91} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

1125. $L_{91} \circ L_{291}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1126. $L_{91} \circ L_{340}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1127. $L_{91} \circ L_{401}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1128. $L_{91} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1129. $L_{91} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1130. $L_{91} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1131. $L_{91} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1132. $L_{91} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1133. $L_{91} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1134. $L_{91} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1135. $L_{91} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1136. $L_{91} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1137. $L_{91} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1138. $L_{91} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

1139. $L_{91} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1140. $L_{91} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1141. $L_{91} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1142. $L_{91} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1143. $L_{91} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

1144. $L_{92} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1145. $L_{92} \circ L_{102}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1146. $L_{92} \circ L_{129}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1147. $L_{92} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1148. $L_{92} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1149. $L_{92} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1150. $L_{92} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1151. $L_{92} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1152. $L_{92} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

1153. $L_{92} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1154. $L_{92} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1155. $L_{92} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1156. $L_{92} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

1157. $L_{92} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

1158. $L_{92} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1159. $L_{92} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1160. $L_{92} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1161. $L_{92} \circ L_{294}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1162. $L_{92} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1163. $L_{92} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1164. $L_{92} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \\ 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

1165. $L_{92} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1166. $L_{92} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1167. $L_{92} \circ L_{401}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1168. $L_{92} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1169. $L_{92} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1170. $L_{92} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1171. $L_{92} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1172. $L_{92} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1173. $L_{92} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1174. $L_{92} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1175. $L_{92} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1176. $L_{92} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1177. $L_{92} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1178. $L_{92} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1179. $L_{92} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

1180. $L_{92} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1181. $L_{92} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1182. $L_{92} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1183. $L_{92} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1184. $L_{93} \circ L_{236}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1185. $L_{93} \circ L_{240}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1186. $L_{93} \circ L_{289}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1187. $L_{93} \circ L_{304}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1188. $L_{93} \circ L_{342}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1189. $L_{93} \circ L_{359}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1190. $L_{93} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1191. $L_{93} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1192. $L_{93} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1193. $L_{93} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1194. $L_{93} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1195. $L_{93} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1196. $L_{93} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1197. $L_{93} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1198. $L_{93} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1199. $L_{93} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (143)(2) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1200. $L_{94} \circ L_{99}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1201. $L_{94} \circ L_{102}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1202. $L_{94} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1203. $L_{94} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1204. $L_{94} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1205. $L_{94} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1206. $L_{94} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1207. $L_{94} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1208. $L_{94} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1209. $L_{94} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1210. $L_{94} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1211. $L_{94} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1212. $L_{94} \circ L_{294}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1213. $L_{94} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1214. $L_{94} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1215. $L_{94} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

1216. $L_{94} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1217. $L_{94} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1218. $L_{94} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1219. $L_{94} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1220. $L_{94} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1221. $L_{94} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1222. $L_{94} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1223. $L_{94} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1224. $L_{94} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1225. $L_{94} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1226. $L_{94} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1227. $L_{94} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1228. $L_{94} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1229. $L_{94} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

1230. $L_{94} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1231. $L_{94} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 5 & 8 \\ 8 & 5 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1232. $L_{94} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1233. $L_{94} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1234. $L_{96} \circ L_{233}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1235. $L_{96} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1236. $L_{96} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1237. $L_{96} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1238. $L_{96} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1239. $L_{96} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1240. $L_{96} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1241. $L_{96} \circ L_{513}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \end{pmatrix}$$

1242. $L_{96} \circ L_{540}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1243. $L_{96} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1244. $L_{96} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1245. $L_{97} \circ L_{135}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 7 \\ 5 & 8 & 7 & 7 \\ 7 & 7 & 8 & 5 \\ 7 & 7 & 5 & 8 \end{pmatrix}$$

1246. $L_{97} \circ L_{145}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1247. $L_{97} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1248. $L_{97} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1249. $L_{97} \circ L_{184}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1250. $L_{97} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 5 \\ 7 & 7 & 5 & 8 \\ 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \end{pmatrix}$$

1251. $L_{97} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1252. $L_{97} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

1253. $L_{97} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1254. $L_{97} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1255. $L_{97} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1256. $L_{97} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1257. $L_{97} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \\ 8 & 5 & 7 & 7 \\ 5 & 8 & 7 & 7 \end{pmatrix}$$

1258. $L_{97} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1259. $L_{97} \circ L_{342}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1260. $L_{97} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1261. $L_{97} \circ L_{384}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1262. $L_{97} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1263. $L_{97} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1264. $L_{97} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1265. $L_{97} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1266. $L_{97} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1267. $L_{97} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1268. $L_{97} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1269. $L_{97} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1270. $L_{97} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1271. $L_{97} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1272. $L_{97} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1273. $L_{97} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1274. $L_{97} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \\ 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \end{pmatrix}$$

1275. $L_{97} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1276. $L_{98} \circ L_{101}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1277. $L_{98} \circ L_{274}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1278. $L_{98} \circ L_{277}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1279. $L_{98} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1280. $L_{98} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1281. $L_{98} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1282. $L_{98} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1283. $L_{98} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1284. $L_{98} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1285. $L_{98} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1286. $L_{98} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1287. $L_{98} \circ L_{488}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1288. $L_{98} \circ L_{493}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1289. $L_{98} \circ L_{568}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1290. $L_{98} \circ L_{569}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1291. $L_{99} \circ L_{131}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1292. $L_{99} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1293. $L_{99} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1294. $L_{99} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1295. $L_{99} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1296. $L_{99} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1297. $L_{99} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 6 & 7 & 5 & 8 \\ 8 & 5 & 7 & 6 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

1298. $L_{99} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1299. $L_{99} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1300. $L_{99} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1301. $L_{99} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1302. $L_{99} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

1303. $L_{99} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

1304. $L_{99} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1305. $L_{99} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1306. $L_{99} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1307. $L_{99} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 5 \\ 5 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

1308. $L_{99} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1309. $L_{99} \circ L_{340}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1310. $L_{99} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1311. $L_{99} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1312. $L_{99} \circ L_{382}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1313. $L_{99} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1314. $L_{99} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1315. $L_{99} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1316. $L_{99} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1317. $L_{99} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1318. $L_{99} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1319. $L_{99} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1320. $L_{99} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1321. $L_{99} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1322. $L_{99} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1323. $L_{99} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1324. $L_{99} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1325. $L_{99} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1326. $L_{99} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1327. $L_{99} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1328. $L_{99} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

1329. $L_{100} \circ L_{275}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1330. $L_{100} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1331. $L_{100} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1332. $L_{100} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1333. $L_{100} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1334. $L_{100} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1335. $L_{100} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1336. $L_{100} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1337. $L_{100} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1338. $L_{100} \circ L_{495}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1339. $L_{100} \circ L_{566}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \\ 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \end{pmatrix}$$

1340. $L_{101} \circ L_{274}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1341. $L_{101} \circ L_{277}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1342. $L_{101} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1343. $L_{101} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1344. $L_{101} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1345. $L_{101} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1346. $L_{101} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1347. $L_{101} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1348. $L_{101} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1349. $L_{101} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1350. $L_{101} \circ L_{488}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1351. $L_{101} \circ L_{493}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1352. $L_{101} \circ L_{568}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1353. $L_{101} \circ L_{569}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (134)(2) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1354. $L_{102} \circ L_{131}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1355. $L_{102} \circ L_{133}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1356. $L_{102} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1357. $L_{102} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1358. $L_{102} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1359. $L_{102} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1360. $L_{102} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1361. $L_{102} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1362. $L_{102} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1363. $L_{102} \circ L_{259}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1364. $L_{102} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1365. $L_{102} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1366. $L_{102} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1367. $L_{102} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 6 \\ 5 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1368. $L_{102} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1369. $L_{102} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1370. $L_{102} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

1371. $L_{102} \circ L_{382}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1372. $L_{102} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1373. $L_{102} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1374. $L_{102} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1375. $L_{102} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1376. $L_{102} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1377. $L_{102} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1378. $L_{102} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1379. $L_{102} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1380. $L_{102} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 5 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1381. $L_{102} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 5 \\ 6 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

1382. $L_{102} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1383. $L_{102} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1384. $L_{102} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

1385. $L_{102} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1386. $L_{102} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1387. $L_{102} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1388. $L_{104} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1389. $L_{104} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1390. $L_{104} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

1391. $L_{104} \circ L_{279}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1392. $L_{104} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1393. $L_{104} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1394. $L_{104} \circ L_{323}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1395. $L_{104} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1396. $L_{104} \circ L_{340}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1397. $L_{104} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1398. $L_{104} \circ L_{379}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1399. $L_{104} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 5 & 6 \\ 6 & 5 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1400. $L_{104} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 5 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1401. $L_{104} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1402. $L_{104} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 5 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1403. $L_{104} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1404. $L_{104} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1405. $L_{104} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1406. $L_{104} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1407. $L_{104} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1408. $L_{104} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1409. $L_{104} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1410. $L_{104} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

1411. $L_{105} \circ L_{123}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1412. $L_{105} \circ L_{128}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1413. $L_{105} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 8 & 7 & 5 & 6 \\ 6 & 5 & 7 & 8 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

1414. $L_{105} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1415. $L_{105} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1416. $L_{105} \circ L_{227}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1417. $L_{105} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1418. $L_{105} \circ L_{254}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1419. $L_{105} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1420. $L_{105} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1421. $L_{105} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1422. $L_{105} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1423. $L_{105} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 5 & 7 & 8 \end{pmatrix}$$

1424. $L_{105} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1425. $L_{105} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1426. $L_{105} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

1427. $L_{105} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1428. $L_{105} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1429. $L_{105} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

1430. $L_{105} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1431. $L_{105} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1432. $L_{105} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1433. $L_{105} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1434. $L_{105} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1435. $L_{105} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1436. $L_{105} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1437. $L_{107} \circ L_{108}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1438. $L_{107} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

1439. $L_{107} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1440. $L_{107} \circ L_{267}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1441. $L_{107} \circ L_{268}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1442. $L_{107} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1443. $L_{107} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1444. $L_{107} \circ L_{313}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1445. $L_{107} \circ L_{315}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1446. $L_{107} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1447. $L_{107} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1448. $L_{107} \circ L_{369}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1449. $L_{107} \circ L_{371}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1450. $L_{107} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

1451. $L_{107} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1452. $L_{107} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1453. $L_{107} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1454. $L_{107} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1455. $L_{107} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1456. $L_{107} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1457. $L_{107} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 5 & 8 \\ 8 & 5 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1458. $L_{108} \circ L_{267}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1459. $L_{108} \circ L_{268}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1460. $L_{108} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1461. $L_{108} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1462. $L_{108} \circ L_{313}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1463. $L_{108} \circ L_{315}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1464. $L_{108} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1465. $L_{108} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1466. $L_{108} \circ L_{369}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1467. $L_{108} \circ L_{371}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1468. $L_{108} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1469. $L_{108} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1470. $L_{108} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1471. $L_{108} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1472. $L_{108} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1473. $L_{108} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1474. $L_{109} \circ L_{110}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1475. $L_{109} \circ L_{269}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1476. $L_{109} \circ L_{270}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (1342) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1477. $L_{110} \circ L_{269}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1478. $L_{110} \circ L_{270}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (132)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1479. $L_{112} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1480. $L_{112} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1481. $L_{114} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (142)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1482. $L_{114} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (142)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1483. $L_{115} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1484. $L_{115} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1485. $L_{115} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1486. $L_{115} \circ L_{206}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

1487. $L_{115} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1488. $L_{115} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1489. $L_{115} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1490. $L_{115} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

1491. $L_{115} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1492. $L_{115} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1493. $L_{115} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1494. $L_{115} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1495. $L_{115} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1496. $L_{115} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1497. $L_{115} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1498. $L_{115} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1499. $L_{115} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1500. $L_{115} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

1501. $L_{116} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1502. $L_{116} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1503. $L_{116} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1504. $L_{116} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1505. $L_{118} \circ L_{120}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1506. $L_{118} \circ L_{281}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1507. $L_{118} \circ L_{283}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(243) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1508. $L_{119} \circ L_{138}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1509. $L_{119} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1510. $L_{119} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1511. $L_{119} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 5 \\ 5 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 5 & 8 \end{pmatrix}$$

1512. $L_{119} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1513. $L_{119} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1514. $L_{119} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1515. $L_{119} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1516. $L_{119} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1517. $L_{119} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1518. $L_{119} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 5 \\ 5 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

1519. $L_{119} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1520. $L_{119} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1521. $L_{119} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1522. $L_{119} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1523. $L_{119} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

1524. $L_{119} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1525. $L_{119} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1526. $L_{119} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

1527. $L_{120} \circ L_{281}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1528. $L_{120} \circ L_{283}$

$$A = \begin{pmatrix} 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1529. $L_{122} \circ L_{127}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1530. $L_{122} \circ L_{251}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1531. $L_{122} \circ L_{255}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1532. $L_{123} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1533. $L_{123} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1534. $L_{123} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1535. $L_{123} \circ L_{227}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

1536. $L_{123} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1537. $L_{123} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1538. $L_{123} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1539. $L_{123} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

1540. $L_{123} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1541. $L_{123} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1542. $L_{123} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1543. $L_{123} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1544. $L_{123} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1545. $L_{123} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1546. $L_{123} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

1547. $L_{123} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 5 & 8 \\ 8 & 5 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1548. $L_{123} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1549. $L_{124} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1550. $L_{124} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1551. $L_{124} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1552. $L_{124} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(234) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

1553. $L_{125} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (132)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1554. $L_{125} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (132)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1555. $L_{127} \circ L_{251}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (132)(4) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (132)(4) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1556. $L_{127} \circ L_{255}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(34) \\ \tau_2 = (1423) \\ \tau_3 = (132)(4) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (132)(4) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1557. $L_{128} \circ L_{159}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1558. $L_{128} \circ L_{176}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1559. $L_{128} \circ L_{216}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1560. $L_{128} \circ L_{232}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 5 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 5 & 6 & 8 \end{pmatrix}$$

1561. $L_{128} \circ L_{256}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1562. $L_{128} \circ L_{265}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1563. $L_{128} \circ L_{303}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1564. $L_{128} \circ L_{320}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1565. $L_{128} \circ L_{360}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 8 & 6 & 5 & 7 \\ 7 & 5 & 6 & 8 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

1566. $L_{128} \circ L_{376}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1567. $L_{128} \circ L_{400}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1568. $L_{128} \circ L_{409}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1569. $L_{128} \circ L_{447}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1570. $L_{128} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1571. $L_{128} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1572. $L_{128} \circ L_{512}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

1573. $L_{128} \circ L_{551}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1574. $L_{128} \circ L_{560}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1575. $L_{129} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

1576. $L_{129} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1577. $L_{129} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 7 \\ 5 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1578. $L_{129} \circ L_{257}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1579. $L_{129} \circ L_{294}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1580. $L_{129} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1581. $L_{129} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1582. $L_{129} \circ L_{337}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1583. $L_{129} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1584. $L_{129} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1585. $L_{129} \circ L_{382}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1586. $L_{129} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

1587. $L_{129} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

1588. $L_{129} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 5 \\ 7 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

1589. $L_{129} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 5 \\ 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

1590. $L_{129} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1591. $L_{129} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1592. $L_{129} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1593. $L_{129} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1594. $L_{129} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1595. $L_{129} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1596. $L_{129} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1597. $L_{129} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1598. $L_{131} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1599. $L_{131} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1600. $L_{131} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1601. $L_{131} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1602. $L_{131} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1603. $L_{131} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1604. $L_{131} \circ L_{235}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1605. $L_{131} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1606. $L_{131} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1607. $L_{131} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

1608. $L_{131} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1609. $L_{131} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1610. $L_{131} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1611. $L_{131} \circ L_{340}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1612. $L_{131} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1613. $L_{131} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1614. $L_{131} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1615. $L_{131} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1616. $L_{131} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1617. $L_{131} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1618. $L_{131} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1619. $L_{131} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1620. $L_{131} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1621. $L_{131} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1622. $L_{131} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1623. $L_{131} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 5 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1624. $L_{131} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1625. $L_{131} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 5 & 7 \\ 7 & 5 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1626. $L_{131} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 5 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1627. $L_{131} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

1628. $L_{131} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1629. $L_{131} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(23)(4) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1630. $L_{132} \circ L_{136}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1631. $L_{132} \circ L_{260}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1632. $L_{132} \circ L_{264}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1633. $L_{132} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1634. $L_{132} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1635. $L_{132} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1636. $L_{132} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1637. $L_{132} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1638. $L_{132} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1639. $L_{132} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1640. $L_{132} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1641. $L_{132} \circ L_{511}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1642. $L_{132} \circ L_{520}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1643. $L_{132} \circ L_{541}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1644. $L_{132} \circ L_{552}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(234) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1645. $L_{133} \circ L_{147}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1646. $L_{133} \circ L_{148}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1647. $L_{133} \circ L_{181}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1648. $L_{133} \circ L_{182}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1649. $L_{133} \circ L_{193}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1650. $L_{133} \circ L_{195}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1651. $L_{133} \circ L_{237}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 5 \\ 5 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

1652. $L_{133} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

1653. $L_{133} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1654. $L_{133} \circ L_{273}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1655. $L_{133} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1656. $L_{133} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

1657. $L_{133} \circ L_{292}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 7 & 6 & 5 & 8 \\ 8 & 5 & 6 & 7 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

1658. $L_{133} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1659. $L_{133} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1660. $L_{133} \circ L_{339}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1661. $L_{133} \circ L_{340}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1662. $L_{133} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1663. $L_{133} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1664. $L_{133} \circ L_{382}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1665. $L_{133} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1666. $L_{133} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1667. $L_{133} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1668. $L_{133} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1669. $L_{133} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1670. $L_{133} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1671. $L_{133} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1672. $L_{133} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1673. $L_{133} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1674. $L_{133} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1675. $L_{133} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1676. $L_{133} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1677. $L_{133} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1678. $L_{133} \circ L_{544}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

1679. $L_{133} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1680. $L_{133} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1681. $L_{134} \circ L_{262}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1682. $L_{134} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1683. $L_{134} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1684. $L_{134} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1685. $L_{134} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1686. $L_{134} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1687. $L_{134} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1688. $L_{134} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1689. $L_{134} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1690. $L_{134} \circ L_{518}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1691. $L_{134} \circ L_{543}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \\ 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \end{pmatrix}$$

1692. $L_{135} \circ L_{145}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1693. $L_{135} \circ L_{146}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1694. $L_{135} \circ L_{183}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1695. $L_{135} \circ L_{184}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1696. $L_{135} \circ L_{197}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1697. $L_{135} \circ L_{239}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \\ 8 & 5 & 7 & 7 \\ 5 & 8 & 7 & 7 \end{pmatrix}$$

1698. $L_{135} \circ L_{261}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1699. $L_{135} \circ L_{263}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

1700. $L_{135} \circ L_{276}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

1701. $L_{135} \circ L_{278}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1702. $L_{135} \circ L_{290}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 5 \\ 7 & 7 & 5 & 8 \\ 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \end{pmatrix}$$

1703. $L_{135} \circ L_{327}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1704. $L_{135} \circ L_{341}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1705. $L_{135} \circ L_{342}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1706. $L_{135} \circ L_{383}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1707. $L_{135} \circ L_{384}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1708. $L_{135} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1709. $L_{135} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1710. $L_{135} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1711. $L_{135} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1712. $L_{135} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1713. $L_{135} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1714. $L_{135} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1715. $L_{135} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1716. $L_{135} \circ L_{494}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1717. $L_{135} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1718. $L_{135} \circ L_{517}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1719. $L_{135} \circ L_{519}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1720. $L_{135} \circ L_{542}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1721. $L_{135} \circ L_{567}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \\ 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \end{pmatrix}$$

1722. $L_{136} \circ L_{260}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1723. $L_{136} \circ L_{264}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1724. $L_{136} \circ L_{405}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1725. $L_{136} \circ L_{407}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1726. $L_{136} \circ L_{420}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1727. $L_{136} \circ L_{422}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1728. $L_{136} \circ L_{434}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1729. $L_{136} \circ L_{436}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1730. $L_{136} \circ L_{469}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1731. $L_{136} \circ L_{471}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1732. $L_{136} \circ L_{511}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1733. $L_{136} \circ L_{520}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1734. $L_{136} \circ L_{541}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1735. $L_{136} \circ L_{552}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1736. $L_{138} \circ L_{167}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1737. $L_{138} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 5 \\ 5 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

1738. $L_{138} \circ L_{202}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1739. $L_{138} \circ L_{206}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1740. $L_{138} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1741. $L_{138} \circ L_{247}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

1742. $L_{138} \circ L_{282}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1743. $L_{138} \circ L_{286}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1744. $L_{138} \circ L_{311}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 5 \\ 5 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 5 & 8 \end{pmatrix}$$

1745. $L_{138} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1746. $L_{138} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1747. $L_{138} \circ L_{346}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

1748. $L_{138} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1749. $L_{138} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1750. $L_{138} \circ L_{391}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1751. $L_{138} \circ L_{426}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1752. $L_{138} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1753. $L_{138} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

1754. $L_{138} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(2)(3)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1755. $L_{138} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1756. $L_{138} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1757. $L_{138} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1758. $L_{138} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1759. $L_{138} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(2)(3)(4) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1760. $L_{139} \circ L_{140}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1761. $L_{139} \circ L_{243}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1762. $L_{139} \circ L_{244}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1763. $L_{140} \circ L_{243}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1764. $L_{140} \circ L_{244}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

1765. $L_{141} \circ L_{142}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1766. $L_{141} \circ L_{241}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1767. $L_{141} \circ L_{242}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1768. $L_{141} \circ L_{307}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1769. $L_{141} \circ L_{309}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1770. $L_{141} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1771. $L_{141} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1772. $L_{141} \circ L_{348}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1773. $L_{141} \circ L_{350}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1774. $L_{141} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1775. $L_{141} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1776. $L_{141} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1777. $L_{141} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1778. $L_{141} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1779. $L_{141} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1780. $L_{141} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1781. $L_{141} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(243) \\ \tau_2 = (142)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1782. $L_{142} \circ L_{186}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1783. $L_{142} \circ L_{223}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

1784. $L_{142} \circ L_{241}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1785. $L_{142} \circ L_{242}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1786. $L_{142} \circ L_{307}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1787. $L_{142} \circ L_{309}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1788. $L_{142} \circ L_{325}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1789. $L_{142} \circ L_{330}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1790. $L_{142} \circ L_{348}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1791. $L_{142} \circ L_{350}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1792. $L_{142} \circ L_{367}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1793. $L_{142} \circ L_{381}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1794. $L_{142} \circ L_{455}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1795. $L_{142} \circ L_{474}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1796. $L_{142} \circ L_{487}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1797. $L_{142} \circ L_{496}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1798. $L_{142} \circ L_{522}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1799. $L_{142} \circ L_{535}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1)(24)(3) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

1800. $L_{142} \circ L_{565}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1801. $L_{142} \circ L_{570}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(24)(3) \\ \tau_2 = (1432) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1802. $L_{143} \circ L_{464}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1803. $L_{143} \circ L_{503}$

$$A = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1)(23)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1804. $L_{145} \circ L_{197}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1805. $L_{145} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1806. $L_{145} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1807. $L_{145} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1808. $L_{145} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1809. $L_{145} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1810. $L_{145} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1811. $L_{145} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1812. $L_{145} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1813. $L_{145} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

1814. $L_{145} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

1815. $L_{145} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1816. $L_{145} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1817. $L_{145} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1818. $L_{145} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1819. $L_{145} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1820. $L_{146} \circ L_{183}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

1821. $L_{146} \circ L_{197}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1822. $L_{146} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1823. $L_{146} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1824. $L_{146} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1825. $L_{146} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1826. $L_{146} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1827. $L_{146} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1828. $L_{146} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1829. $L_{146} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

1830. $L_{146} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1831. $L_{146} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1832. $L_{146} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

1833. $L_{146} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1834. $L_{146} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1835. $L_{146} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1836. $L_{146} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1837. $L_{147} \circ L_{193}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1838. $L_{147} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1839. $L_{147} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1840. $L_{147} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1841. $L_{147} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1842. $L_{147} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1843. $L_{147} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1844. $L_{147} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1845. $L_{147} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1846. $L_{147} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1847. $L_{147} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1848. $L_{147} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1849. $L_{147} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1850. $L_{147} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1851. $L_{147} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1852. $L_{147} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1853. $L_{147} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

1854. $L_{147} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1855. $L_{147} \circ L_{423}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1856. $L_{147} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1857. $L_{147} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1858. $L_{147} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

1859. $L_{147} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1860. $L_{147} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1861. $L_{147} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1862. $L_{147} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1863. $L_{147} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1864. $L_{147} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1865. $L_{147} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1866. $L_{147} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1867. $L_{147} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1868. $L_{148} \circ L_{159}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \end{pmatrix}$$

1869. $L_{148} \circ L_{179}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

1870. $L_{148} \circ L_{181}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

1871. $L_{148} \circ L_{193}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1872. $L_{148} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1873. $L_{148} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 5 \\ 5 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1874. $L_{148} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1875. $L_{148} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1876. $L_{148} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1877. $L_{148} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1878. $L_{148} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1879. $L_{148} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1880. $L_{148} \circ L_{279}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1881. $L_{148} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1882. $L_{148} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1883. $L_{148} \circ L_{323}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 5 & 6 \\ 6 & 5 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1884. $L_{148} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1885. $L_{148} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1886. $L_{148} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1887. $L_{148} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1888. $L_{148} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1889. $L_{148} \circ L_{379}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1890. $L_{148} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1891. $L_{148} \circ L_{403}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1892. $L_{148} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

1893. $L_{148} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

1894. $L_{148} \circ L_{423}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

1895. $L_{148} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

1896. $L_{148} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

1897. $L_{148} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1898. $L_{148} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1899. $L_{148} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1900. $L_{148} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1901. $L_{148} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1902. $L_{148} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1903. $L_{148} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1904. $L_{148} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1905. $L_{148} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1906. $L_{148} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1907. $L_{148} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1908. $L_{148} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1909. $L_{149} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (134)(2) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1910. $L_{149} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (134)(2) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1911. $L_{150} \circ L_{179}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1912. $L_{150} \circ L_{181}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

1913. $L_{150} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

1914. $L_{150} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1915. $L_{150} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1916. $L_{150} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1917. $L_{150} \circ L_{379}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1918. $L_{150} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1919. $L_{150} \circ L_{401}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

1920. $L_{150} \circ L_{417}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \end{pmatrix}$$

1921. $L_{150} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1922. $L_{150} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1923. $L_{150} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1924. $L_{150} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1925. $L_{150} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1926. $L_{150} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

1927. $L_{150} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1928. $L_{150} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

1929. $L_{152} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1930. $L_{152} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1931. $L_{152} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1932. $L_{152} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1933. $L_{152} \circ L_{439}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1934. $L_{152} \circ L_{467}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1935. $L_{152} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1936. $L_{152} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1937. $L_{152} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1938. $L_{152} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1939. $L_{153} \circ L_{156}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1940. $L_{153} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1941. $L_{153} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1942. $L_{153} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1943. $L_{153} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1944. $L_{153} \circ L_{444}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1945. $L_{153} \circ L_{450}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1946. $L_{153} \circ L_{459}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1947. $L_{153} \circ L_{479}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1948. $L_{153} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1949. $L_{153} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (1342) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1950. $L_{153} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1951. $L_{153} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1952. $L_{154} \circ L_{157}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (143)(2) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (143)(2) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1953. $L_{156} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1954. $L_{156} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1955. $L_{156} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1956. $L_{156} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1957. $L_{156} \circ L_{444}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1958. $L_{156} \circ L_{450}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1959. $L_{156} \circ L_{459}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1960. $L_{156} \circ L_{479}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1961. $L_{156} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1962. $L_{156} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1963. $L_{156} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1964. $L_{156} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1965. $L_{159} \circ L_{171}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1966. $L_{159} \circ L_{176}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

1967. $L_{159} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1968. $L_{159} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

1969. $L_{159} \circ L_{227}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1970. $L_{159} \circ L_{232}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

1971. $L_{159} \circ L_{254}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1972. $L_{159} \circ L_{256}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

1973. $L_{159} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1974. $L_{159} \circ L_{267}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

1975. $L_{159} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1976. $L_{159} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1977. $L_{159} \circ L_{315}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

1978. $L_{159} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

1979. $L_{159} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1980. $L_{159} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

1981. $L_{159} \circ L_{371}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1982. $L_{159} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

1983. $L_{159} \circ L_{398}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1984. $L_{159} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1985. $L_{159} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1986. $L_{159} \circ L_{411}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 5 & 6 \\ 6 & 5 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

1987. $L_{159} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

1988. $L_{159} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 7 & 6 & 8 & 5 \\ 5 & 8 & 6 & 7 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

1989. $L_{159} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1990. $L_{159} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

1991. $L_{159} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1992. $L_{159} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1993. $L_{159} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1994. $L_{159} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1995. $L_{159} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1996. $L_{159} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

1997. $L_{159} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1998. $L_{159} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 5 \\ 5 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

1999. $L_{160} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2000. $L_{160} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2001. $L_{160} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

2002. $L_{160} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2003. $L_{161} \circ L_{163}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2004. $L_{161} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2005. $L_{161} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2006. $L_{161} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2007. $L_{161} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2008. $L_{161} \circ L_{386}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2009. $L_{161} \circ L_{393}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2010. $L_{161} \circ L_{416}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2011. $L_{161} \circ L_{430}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2012. $L_{161} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2013. $L_{161} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2014. $L_{161} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2015. $L_{161} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2016. $L_{161} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2017. $L_{161} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2018. $L_{162} \circ L_{166}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (142)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (134)(2) \\ \tau_4 = (142)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2019. $L_{163} \circ L_{186}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2020. $L_{163} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2021. $L_{163} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2022. $L_{163} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2023. $L_{163} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2024. $L_{163} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2025. $L_{163} \circ L_{386}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2026. $L_{163} \circ L_{393}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2027. $L_{163} \circ L_{416}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2028. $L_{163} \circ L_{430}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2029. $L_{163} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2030. $L_{163} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2031. $L_{163} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2032. $L_{163} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2033. $L_{163} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2034. $L_{163} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2035. $L_{165} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2036. $L_{165} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2037. $L_{165} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2038. $L_{165} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

2039. $L_{167} \circ L_{186}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2040. $L_{167} \circ L_{202}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2041. $L_{167} \circ L_{206}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2042. $L_{167} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2043. $L_{167} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \end{pmatrix}$$

2044. $L_{167} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2045. $L_{167} \circ L_{286}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2046. $L_{167} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2047. $L_{167} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2048. $L_{167} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2049. $L_{167} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \end{pmatrix}$$

2050. $L_{167} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2051. $L_{167} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2052. $L_{167} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

2053. $L_{167} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

2054. $L_{167} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \end{pmatrix}$$

2055. $L_{167} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2056. $L_{167} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2057. $L_{167} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2058. $L_{167} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2059. $L_{167} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2060. $L_{167} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2061. $L_{167} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2062. $L_{169} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2063. $L_{169} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2064. $L_{169} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2065. $L_{169} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2066. $L_{170} \circ L_{173}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2067. $L_{171} \circ L_{174}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2068. $L_{171} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2069. $L_{171} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2070. $L_{171} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2071. $L_{171} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2072. $L_{171} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2073. $L_{171} \circ L_{390}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2074. $L_{171} \circ L_{398}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2075. $L_{171} \circ L_{411}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2076. $L_{171} \circ L_{431}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2077. $L_{171} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2078. $L_{171} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2079. $L_{171} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2080. $L_{171} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2081. $L_{171} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2082. $L_{171} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2083. $L_{174} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2084. $L_{174} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2085. $L_{174} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2086. $L_{174} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2087. $L_{174} \circ L_{390}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2088. $L_{174} \circ L_{398}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2089. $L_{174} \circ L_{411}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2090. $L_{174} \circ L_{431}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2091. $L_{174} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 5 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2092. $L_{174} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2093. $L_{174} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2094. $L_{174} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2095. $L_{174} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2096. $L_{174} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

2097. $L_{176} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2098. $L_{176} \circ L_{227}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2099. $L_{176} \circ L_{232}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2100. $L_{176} \circ L_{254}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2101. $L_{176} \circ L_{256}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2102. $L_{176} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 5 \\ 7 & 8 & 5 & 7 \\ 7 & 5 & 8 & 7 \\ 5 & 7 & 7 & 8 \end{pmatrix}$$

2103. $L_{176} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2104. $L_{176} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2105. $L_{176} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2106. $L_{176} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2107. $L_{176} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2108. $L_{176} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 7 & 8 \\ 7 & 5 & 8 & 7 \\ 7 & 8 & 5 & 7 \\ 8 & 7 & 7 & 5 \end{pmatrix}$$

2109. $L_{176} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

2110. $L_{176} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

2111. $L_{176} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2112. $L_{176} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 7 \\ 8 & 7 & 7 & 5 \\ 5 & 7 & 7 & 8 \\ 7 & 8 & 5 & 7 \end{pmatrix}$$

2113. $L_{176} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2114. $L_{176} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2115. $L_{176} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2116. $L_{176} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2117. $L_{176} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2118. $L_{176} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2119. $L_{177} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2120. $L_{177} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2121. $L_{177} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2122. $L_{177} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2123. $L_{177} \circ L_{437}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2124. $L_{177} \circ L_{465}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2125. $L_{177} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2126. $L_{177} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2127. $L_{177} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2128. $L_{177} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2129. $L_{179} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2130. $L_{179} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2131. $L_{179} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2132. $L_{179} \circ L_{337}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2133. $L_{179} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2134. $L_{179} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2135. $L_{179} \circ L_{403}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 7 & 8 \\ 7 & 5 & 8 & 7 \\ 7 & 8 & 5 & 7 \\ 8 & 7 & 7 & 5 \end{pmatrix}$$

2136. $L_{179} \circ L_{423}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2137. $L_{179} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2138. $L_{179} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2139. $L_{179} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2140. $L_{179} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2141. $L_{179} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2142. $L_{179} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2143. $L_{179} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2144. $L_{179} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2145. $L_{180} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (143)(2) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2146. $L_{180} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (143)(2) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2147. $L_{181} \circ L_{186}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2148. $L_{181} \circ L_{193}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2149. $L_{181} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2150. $L_{181} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \end{pmatrix}$$

2151. $L_{181} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2152. $L_{181} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2153. $L_{181} \circ L_{257}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2154. $L_{181} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2155. $L_{181} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2156. $L_{181} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2157. $L_{181} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2158. $L_{181} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2159. $L_{181} \circ L_{294}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 5 \\ 7 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

2160. $L_{181} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2161. $L_{181} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2162. $L_{181} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2163. $L_{181} \circ L_{337}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2164. $L_{181} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2165. $L_{181} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2166. $L_{181} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2167. $L_{181} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2168. $L_{181} \circ L_{401}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2169. $L_{181} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

2170. $L_{181} \circ L_{417}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2171. $L_{181} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2172. $L_{181} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2173. $L_{181} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2174. $L_{181} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

2175. $L_{181} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2176. $L_{181} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2177. $L_{181} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2178. $L_{181} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2179. $L_{181} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2180. $L_{181} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2181. $L_{181} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

2182. $L_{181} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2183. $L_{181} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2184. $L_{181} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2185. $L_{181} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2186. $L_{182} \circ L_{193}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2187. $L_{182} \circ L_{195}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2188. $L_{182} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2189. $L_{182} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2190. $L_{182} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2191. $L_{182} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2192. $L_{182} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2193. $L_{182} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2194. $L_{182} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2195. $L_{182} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2196. $L_{182} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2197. $L_{182} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2198. $L_{182} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2199. $L_{182} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2200. $L_{182} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2201. $L_{182} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2202. $L_{182} \circ L_{401}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2203. $L_{182} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2204. $L_{182} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

2205. $L_{182} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

2206. $L_{182} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2207. $L_{182} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2208. $L_{182} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2209. $L_{182} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2210. $L_{182} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2211. $L_{182} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2212. $L_{182} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2213. $L_{182} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2214. $L_{182} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2215. $L_{182} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2216. $L_{182} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2217. $L_{183} \circ L_{197}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2218. $L_{183} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2219. $L_{183} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2220. $L_{183} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2221. $L_{183} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2222. $L_{183} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2223. $L_{183} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2224. $L_{183} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2225. $L_{183} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2226. $L_{183} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

2227. $L_{183} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

2228. $L_{183} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2229. $L_{183} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2230. $L_{183} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2231. $L_{183} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2232. $L_{183} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2233. $L_{184} \circ L_{197}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2234. $L_{184} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2235. $L_{184} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2236. $L_{184} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2237. $L_{184} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2238. $L_{184} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2239. $L_{184} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2240. $L_{184} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2241. $L_{184} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

2242. $L_{184} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2243. $L_{184} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2244. $L_{184} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

2245. $L_{184} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2246. $L_{184} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2247. $L_{184} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2248. $L_{184} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2249. $L_{185} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (142)(3) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2250. $L_{185} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (142)(3) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

2251. $L_{185} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (142)(3) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2252. $L_{185} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (142)(3) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2253. $L_{186} \circ L_{202}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2254. $L_{186} \circ L_{206}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2255. $L_{186} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2256. $L_{186} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \end{pmatrix}$$

2257. $L_{186} \circ L_{242}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2258. $L_{186} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2259. $L_{186} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2260. $L_{186} \circ L_{286}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2261. $L_{186} \circ L_{307}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2262. $L_{186} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

2263. $L_{186} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2264. $L_{186} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2265. $L_{186} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2266. $L_{186} \circ L_{350}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2267. $L_{186} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2268. $L_{186} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2269. $L_{186} \circ L_{386}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

2270. $L_{186} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2271. $L_{186} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2272. $L_{186} \circ L_{430}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2273. $L_{186} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2274. $L_{186} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2275. $L_{186} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2276. $L_{186} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 5 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 5 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2277. $L_{186} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2278. $L_{186} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2279. $L_{186} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2280. $L_{186} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2281. $L_{186} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2282. $L_{186} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2283. $L_{186} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2284. $L_{186} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2285. $L_{187} \circ L_{191}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2286. $L_{190} \circ L_{192}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2287. $L_{190} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2288. $L_{190} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2289. $L_{190} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2290. $L_{190} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2291. $L_{190} \circ L_{442}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2292. $L_{190} \circ L_{451}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2293. $L_{190} \circ L_{461}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2294. $L_{190} \circ L_{478}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2295. $L_{190} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2296. $L_{190} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2297. $L_{190} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2298. $L_{190} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2299. $L_{192} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2300. $L_{192} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2301. $L_{192} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2302. $L_{192} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2303. $L_{192} \circ L_{442}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2304. $L_{192} \circ L_{451}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2305. $L_{192} \circ L_{461}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2306. $L_{192} \circ L_{478}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2307. $L_{192} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2308. $L_{192} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2309. $L_{192} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (1)(234) \\ \tau_3 = (1432) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2310. $L_{192} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1432) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2311. $L_{193} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2312. $L_{193} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2313. $L_{193} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2314. $L_{193} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2315. $L_{193} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2316. $L_{193} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2317. $L_{193} \circ L_{323}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2318. $L_{193} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2319. $L_{193} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2320. $L_{193} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2321. $L_{193} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 5 \\ 6 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

2322. $L_{193} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2323. $L_{193} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2324. $L_{193} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2325. $L_{193} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2326. $L_{193} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2327. $L_{193} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2328. $L_{193} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 5 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2329. $L_{193} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2330. $L_{193} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2331. $L_{193} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

2332. $L_{193} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2333. $L_{193} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2334. $L_{193} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2335. $L_{193} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2336. $L_{193} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2337. $L_{193} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 6 \\ 5 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2338. $L_{194} \circ L_{198}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2339. $L_{194} \circ L_{328}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2340. $L_{194} \circ L_{329}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2341. $L_{194} \circ L_{368}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2342. $L_{194} \circ L_{384}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2343. $L_{194} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2344. $L_{194} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2345. $L_{194} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2346. $L_{194} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2347. $L_{194} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2348. $L_{194} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2349. $L_{194} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2350. $L_{194} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2351. $L_{194} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

2352. $L_{194} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2353. $L_{195} \circ L_{216}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 5 & 8 \end{pmatrix}$$

2354. $L_{195} \circ L_{235}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2355. $L_{195} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2356. $L_{195} \circ L_{259}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2357. $L_{195} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2358. $L_{195} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

2359. $L_{195} \circ L_{279}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

2360. $L_{195} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2361. $L_{195} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2362. $L_{195} \circ L_{323}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

2363. $L_{195} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

2364. $L_{195} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2365. $L_{195} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2366. $L_{195} \circ L_{379}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2367. $L_{195} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2368. $L_{195} \circ L_{403}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2369. $L_{195} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2370. $L_{195} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2371. $L_{195} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2372. $L_{195} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2373. $L_{195} \circ L_{423}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 5 & 8 \\ 8 & 5 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2374. $L_{195} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2375. $L_{195} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2376. $L_{195} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 5 \\ 5 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2377. $L_{195} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2378. $L_{195} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2379. $L_{195} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2380. $L_{195} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2381. $L_{195} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2382. $L_{195} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2383. $L_{195} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2384. $L_{195} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2385. $L_{195} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2386. $L_{195} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

2387. $L_{195} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 5 \\ 5 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

2388. $L_{196} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2389. $L_{196} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2390. $L_{196} \circ L_{279}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2391. $L_{196} \circ L_{326}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \\ 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \end{pmatrix}$$

2392. $L_{196} \circ L_{382}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2393. $L_{196} \circ L_{403}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2394. $L_{196} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2395. $L_{196} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2396. $L_{196} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2397. $L_{196} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2398. $L_{196} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2399. $L_{196} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2400. $L_{196} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2401. $L_{196} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2402. $L_{196} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2403. $L_{196} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2404. $L_{196} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2405. $L_{196} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2406. $L_{197} \circ L_{239}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 7 \\ 5 & 8 & 7 & 7 \\ 7 & 7 & 8 & 5 \\ 7 & 7 & 5 & 8 \end{pmatrix}$$

2407. $L_{197} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2408. $L_{197} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2409. $L_{197} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \\ 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \end{pmatrix}$$

2410. $L_{197} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

2411. $L_{197} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2412. $L_{197} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

2413. $L_{197} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2414. $L_{197} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2415. $L_{197} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2416. $L_{197} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2417. $L_{197} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2418. $L_{197} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2419. $L_{197} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2420. $L_{197} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2421. $L_{197} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2422. $L_{197} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2423. $L_{197} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2424. $L_{197} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \\ 8 & 5 & 7 & 7 \\ 5 & 8 & 7 & 7 \end{pmatrix}$$

2425. $L_{198} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2426. $L_{198} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2427. $L_{198} \circ L_{328}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2428. $L_{198} \circ L_{329}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2429. $L_{198} \circ L_{368}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2430. $L_{198} \circ L_{384}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2431. $L_{198} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2432. $L_{198} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2433. $L_{198} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2434. $L_{198} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2435. $L_{198} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2436. $L_{198} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2437. $L_{198} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2438. $L_{198} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2439. $L_{198} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

2440. $L_{198} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2441. $L_{199} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2442. $L_{199} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2443. $L_{199} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2444. $L_{199} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2445. $L_{199} \circ L_{490}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2446. $L_{199} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2447. $L_{199} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2448. $L_{199} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2449. $L_{199} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2450. $L_{199} \circ L_{563}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2451. $L_{201} \circ L_{203}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (123)(4) \\ \tau_4 = (142)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (123)(4) \\ \tau_4 = (142)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2452. $L_{202} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2453. $L_{202} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2454. $L_{202} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

2455. $L_{202} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2456. $L_{202} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2457. $L_{202} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2458. $L_{202} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

2459. $L_{202} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2460. $L_{202} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2461. $L_{202} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2462. $L_{202} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2463. $L_{202} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 5 \\ 5 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

2464. $L_{202} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2465. $L_{202} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2466. $L_{202} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2467. $L_{204} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2468. $L_{204} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2469. $L_{204} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2470. $L_{204} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2471. $L_{206} \circ L_{223}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2472. $L_{206} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2473. $L_{206} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2474. $L_{206} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

2475. $L_{206} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2476. $L_{206} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2477. $L_{206} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2478. $L_{206} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2479. $L_{206} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2480. $L_{206} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2481. $L_{206} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2482. $L_{206} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2483. $L_{206} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2484. $L_{206} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (1234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

2485. $L_{208} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2486. $L_{208} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2487. $L_{211} \circ L_{212}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (143)(2) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (143)(2) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2488. $L_{213} \circ L_{214}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2489. $L_{213} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1243) \\ \tau_4 &= (14)(2)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2490. $L_{213} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2491. $L_{213} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2492. $L_{213} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2493. $L_{213} \circ L_{499}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2494. $L_{213} \circ L_{501}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2495. $L_{213} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2496. $L_{213} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2497. $L_{213} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2498. $L_{213} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2499. $L_{213} \circ L_{553}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2500. $L_{213} \circ L_{555}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2501. $L_{214} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2502. $L_{214} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2503. $L_{214} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2504. $L_{214} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2505. $L_{214} \circ L_{499}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2506. $L_{214} \circ L_{501}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2507. $L_{214} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2508. $L_{214} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2509. $L_{214} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2510. $L_{214} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2511. $L_{214} \circ L_{553}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2512. $L_{214} \circ L_{555}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2513. $L_{215} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (124)(3) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

2514. $L_{215} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (124)(3) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2515. $L_{215} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (124)(3) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2516. $L_{215} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (124)(3) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2517. $L_{216} \circ L_{227}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2518. $L_{216} \circ L_{232}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2519. $L_{216} \circ L_{254}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2520. $L_{216} \circ L_{256}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2521. $L_{216} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2522. $L_{216} \circ L_{267}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2523. $L_{216} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2524. $L_{216} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2525. $L_{216} \circ L_{315}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 5 & 8 \\ 8 & 5 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2526. $L_{216} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2527. $L_{216} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2528. $L_{216} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2529. $L_{216} \circ L_{371}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2530. $L_{216} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2531. $L_{216} \circ L_{398}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2532. $L_{216} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2533. $L_{216} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

2534. $L_{216} \circ L_{411}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 5 \end{pmatrix}$$

2535. $L_{216} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 5 & 6 & 8 & 7 \\ 7 & 8 & 6 & 5 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2536. $L_{216} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2537. $L_{216} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2538. $L_{216} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2539. $L_{216} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2540. $L_{216} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2541. $L_{216} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2542. $L_{216} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2543. $L_{216} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2544. $L_{216} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2545. $L_{216} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 5 & 6 & 8 & 7 \\ 7 & 8 & 6 & 5 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

2546. $L_{216} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

2547. $L_{219} \circ L_{220}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2548. $L_{219} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2549. $L_{219} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2550. $L_{219} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2551. $L_{219} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2552. $L_{219} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2553. $L_{219} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2554. $L_{219} \circ L_{524}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2555. $L_{219} \circ L_{526}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2556. $L_{219} \circ L_{530}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2557. $L_{219} \circ L_{532}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2558. $L_{219} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2559. $L_{219} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2560. $L_{220} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2561. $L_{220} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2562. $L_{220} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2563. $L_{220} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2564. $L_{220} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2565. $L_{220} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2566. $L_{220} \circ L_{524}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2567. $L_{220} \circ L_{526}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2568. $L_{220} \circ L_{530}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2569. $L_{220} \circ L_{532}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2570. $L_{220} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2571. $L_{220} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (1234) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2572. $L_{221} \circ L_{222}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (123)(4) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2573. $L_{223} \circ L_{237}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2574. $L_{223} \circ L_{242}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2575. $L_{223} \circ L_{247}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2576. $L_{223} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2577. $L_{223} \circ L_{286}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2578. $L_{223} \circ L_{307}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

2579. $L_{223} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1234) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2580. $L_{223} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2581. $L_{223} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2582. $L_{223} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2583. $L_{223} \circ L_{350}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2584. $L_{223} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2585. $L_{223} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2586. $L_{223} \circ L_{386}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2587. $L_{223} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

2588. $L_{223} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2589. $L_{223} \circ L_{430}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2590. $L_{223} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2591. $L_{223} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2592. $L_{223} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

2593. $L_{223} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2594. $L_{223} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2595. $L_{223} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2596. $L_{223} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2597. $L_{223} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2598. $L_{223} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 5 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 5 & 6 \end{pmatrix}$$

2599. $L_{223} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2600. $L_{223} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2601. $L_{223} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2602. $L_{224} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2603. $L_{224} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2604. $L_{224} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2605. $L_{224} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2606. $L_{225} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2607. $L_{225} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2608. $L_{225} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2609. $L_{225} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

2610. $L_{226} \circ L_{231}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (124)(3) \\ \tau_4 &= (132)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2611. $L_{227} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2612. $L_{227} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 5 & 8 \\ 8 & 5 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2613. $L_{227} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2614. $L_{227} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

2615. $L_{227} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2616. $L_{227} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2617. $L_{227} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2618. $L_{227} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2619. $L_{227} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2620. $L_{227} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2621. $L_{227} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2622. $L_{227} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2623. $L_{227} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1243) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2624. $L_{230} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (123)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

2625. $L_{230} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (123)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2626. $L_{232} \circ L_{256}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1342) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

2627. $L_{232} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2628. $L_{232} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2629. $L_{232} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2630. $L_{232} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

2631. $L_{232} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2632. $L_{232} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2633. $L_{232} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2634. $L_{232} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2635. $L_{232} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2636. $L_{232} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2637. $L_{232} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 8 & 6 & 5 & 7 \\ 7 & 5 & 6 & 8 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2638. $L_{232} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2639. $L_{232} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2640. $L_{233} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2641. $L_{233} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2642. $L_{233} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2643. $L_{233} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2644. $L_{233} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2645. $L_{233} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2646. $L_{233} \circ L_{513}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2647. $L_{233} \circ L_{540}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2648. $L_{233} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2649. $L_{233} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2650. $L_{235} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2651. $L_{235} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2652. $L_{235} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

2653. $L_{235} \circ L_{294}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2654. $L_{235} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2655. $L_{235} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2656. $L_{235} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 5 & 7 \\ 7 & 5 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2657. $L_{235} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2658. $L_{235} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2659. $L_{235} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2660. $L_{235} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2661. $L_{235} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2662. $L_{235} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2663. $L_{235} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2664. $L_{235} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2665. $L_{235} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2666. $L_{235} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2667. $L_{235} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2668. $L_{235} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2669. $L_{235} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2670. $L_{235} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2671. $L_{235} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2672. $L_{235} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2673. $L_{235} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2674. $L_{235} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2675. $L_{236} \circ L_{240}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2676. $L_{236} \circ L_{289}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2677. $L_{236} \circ L_{304}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2678. $L_{236} \circ L_{342}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2679. $L_{236} \circ L_{359}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2680. $L_{236} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2681. $L_{236} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2682. $L_{236} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2683. $L_{236} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2684. $L_{236} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2685. $L_{236} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2686. $L_{236} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2687. $L_{236} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2688. $L_{236} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2689. $L_{236} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

2690. $L_{237} \circ L_{257}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2691. $L_{237} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2692. $L_{237} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2693. $L_{237} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2694. $L_{237} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

2695. $L_{237} \circ L_{294}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2696. $L_{237} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2697. $L_{237} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2698. $L_{237} \circ L_{337}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2699. $L_{237} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2700. $L_{237} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2701. $L_{237} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2702. $L_{237} \circ L_{401}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 5 \\ 7 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

2703. $L_{237} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2704. $L_{237} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2705. $L_{237} \circ L_{417}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2706. $L_{237} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2707. $L_{237} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2708. $L_{237} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2709. $L_{237} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2710. $L_{237} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2711. $L_{237} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2712. $L_{237} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2713. $L_{237} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2714. $L_{237} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2715. $L_{237} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2716. $L_{237} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2717. $L_{237} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2718. $L_{237} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2719. $L_{237} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 5 \\ 7 & 6 & 5 & 8 \\ 8 & 5 & 6 & 7 \\ 5 & 8 & 7 & 6 \end{pmatrix}$$

2720. $L_{237} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2721. $L_{237} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2722. $L_{238} \circ L_{257}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2723. $L_{238} \circ L_{261}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2724. $L_{238} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2725. $L_{238} \circ L_{291}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \\ 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \end{pmatrix}$$

2726. $L_{238} \circ L_{340}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2727. $L_{238} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2728. $L_{238} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2729. $L_{238} \circ L_{417}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2730. $L_{238} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2731. $L_{238} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2732. $L_{238} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2733. $L_{238} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2734. $L_{238} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2735. $L_{238} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2736. $L_{238} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2737. $L_{238} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

2738. $L_{238} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2739. $L_{238} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2740. $L_{239} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2741. $L_{239} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

2742. $L_{239} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

2743. $L_{239} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \\ 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \end{pmatrix}$$

2744. $L_{239} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

2745. $L_{239} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2746. $L_{239} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2747. $L_{239} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2748. $L_{239} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2749. $L_{239} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

2750. $L_{239} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2751. $L_{239} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2752. $L_{239} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2753. $L_{239} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2754. $L_{239} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2755. $L_{239} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2756. $L_{239} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 5 \\ 7 & 7 & 5 & 8 \\ 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \end{pmatrix}$$

2757. $L_{239} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

2758. $L_{240} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2759. $L_{240} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2760. $L_{240} \circ L_{289}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2761. $L_{240} \circ L_{304}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2762. $L_{240} \circ L_{342}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2763. $L_{240} \circ L_{359}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2764. $L_{240} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2765. $L_{240} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2766. $L_{240} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2767. $L_{240} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2768. $L_{240} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2769. $L_{240} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2770. $L_{240} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2771. $L_{240} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2772. $L_{240} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2773. $L_{240} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2774. $L_{241} \circ L_{242}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2775. $L_{241} \circ L_{307}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2776. $L_{241} \circ L_{309}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2777. $L_{241} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2778. $L_{241} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2779. $L_{241} \circ L_{348}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2780. $L_{241} \circ L_{350}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2781. $L_{241} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2782. $L_{241} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2783. $L_{241} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2784. $L_{241} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2785. $L_{241} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2786. $L_{241} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

2787. $L_{241} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2788. $L_{241} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2789. $L_{242} \circ L_{307}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2790. $L_{242} \circ L_{309}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2791. $L_{242} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2792. $L_{242} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2793. $L_{242} \circ L_{348}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2794. $L_{242} \circ L_{350}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2795. $L_{242} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2796. $L_{242} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2797. $L_{242} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2798. $L_{242} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2799. $L_{242} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2800. $L_{242} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2801. $L_{242} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2802. $L_{242} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2803. $L_{243} \circ L_{244}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(243) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2804. $L_{246} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2805. $L_{246} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2806. $L_{246} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2807. $L_{246} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2808. $L_{247} \circ L_{282}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2809. $L_{247} \circ L_{286}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2810. $L_{247} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

2811. $L_{247} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2812. $L_{247} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2813. $L_{247} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

2814. $L_{247} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2815. $L_{247} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2816. $L_{247} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2817. $L_{247} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \end{pmatrix}$$

2818. $L_{247} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2819. $L_{247} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

2820. $L_{247} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2821. $L_{247} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2822. $L_{247} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \end{pmatrix}$$

2823. $L_{247} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2824. $L_{247} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2825. $L_{247} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2826. $L_{249} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2827. $L_{249} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2828. $L_{249} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2829. $L_{249} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2830. $L_{251} \circ L_{255}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2831. $L_{253} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2832. $L_{253} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2833. $L_{254} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2834. $L_{254} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

2835. $L_{254} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2836. $L_{254} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 6 & 5 & 7 \\ 7 & 5 & 6 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2837. $L_{254} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2838. $L_{254} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2839. $L_{254} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2840. $L_{254} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2841. $L_{254} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2842. $L_{254} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2843. $L_{254} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2844. $L_{254} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

2845. $L_{254} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2846. $L_{256} \circ L_{265}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2847. $L_{256} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

2848. $L_{256} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2849. $L_{256} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2850. $L_{256} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2851. $L_{256} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2852. $L_{256} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2853. $L_{256} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2854. $L_{256} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2855. $L_{256} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2856. $L_{256} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2857. $L_{256} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 7 & 6 & 5 & 8 \\ 8 & 5 & 6 & 7 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

2858. $L_{256} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1342) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2859. $L_{257} \circ L_{294}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

2860. $L_{257} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2861. $L_{257} \circ L_{326}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

2862. $L_{257} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2863. $L_{257} \circ L_{337}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \end{pmatrix}$$

2864. $L_{257} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2865. $L_{257} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

2866. $L_{257} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

2867. $L_{257} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2868. $L_{257} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2869. $L_{257} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2870. $L_{257} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2871. $L_{257} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2872. $L_{257} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2873. $L_{257} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2874. $L_{259} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2875. $L_{259} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2876. $L_{259} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 5 & 8 \\ 8 & 5 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2877. $L_{259} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2878. $L_{259} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2879. $L_{259} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

2880. $L_{259} \circ L_{340}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2881. $L_{259} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2882. $L_{259} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2883. $L_{259} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2884. $L_{259} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2885. $L_{259} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2886. $L_{259} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2887. $L_{259} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2888. $L_{259} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2889. $L_{259} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2890. $L_{259} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2891. $L_{259} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2892. $L_{259} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 6 & 8 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

2893. $L_{259} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2894. $L_{259} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

2895. $L_{259} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 8 & 6 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2896. $L_{259} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2897. $L_{259} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2898. $L_{259} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2899. $L_{260} \circ L_{264}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2900. $L_{260} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2901. $L_{260} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2902. $L_{260} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2903. $L_{260} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2904. $L_{260} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2905. $L_{260} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2906. $L_{260} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2907. $L_{260} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2908. $L_{260} \circ L_{511}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2909. $L_{260} \circ L_{520}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2910. $L_{260} \circ L_{541}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2911. $L_{260} \circ L_{552}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (1)(234) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2912. $L_{261} \circ L_{263}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2913. $L_{261} \circ L_{273}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2914. $L_{261} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2915. $L_{261} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \end{pmatrix}$$

2916. $L_{261} \circ L_{291}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2917. $L_{261} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2918. $L_{261} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2919. $L_{261} \circ L_{326}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2920. $L_{261} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2921. $L_{261} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

2922. $L_{261} \circ L_{340}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2923. $L_{261} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2924. $L_{261} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2925. $L_{261} \circ L_{382}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

2926. $L_{261} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2927. $L_{261} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2928. $L_{261} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2929. $L_{261} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2930. $L_{261} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2931. $L_{261} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2932. $L_{261} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2933. $L_{261} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2934. $L_{261} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2935. $L_{261} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

2936. $L_{261} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2937. $L_{261} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \\ 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

2938. $L_{261} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2939. $L_{261} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2940. $L_{261} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2941. $L_{261} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2942. $L_{261} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

2943. $L_{261} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2944. $L_{262} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2945. $L_{262} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2946. $L_{262} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2947. $L_{262} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2948. $L_{262} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2949. $L_{262} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2950. $L_{262} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2951. $L_{262} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2952. $L_{262} \circ L_{518}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2953. $L_{262} \circ L_{543}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2954. $L_{263} \circ L_{276}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \end{pmatrix}$$

2955. $L_{263} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

2956. $L_{263} \circ L_{289}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

2957. $L_{263} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2958. $L_{263} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

2959. $L_{263} \circ L_{328}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2960. $L_{263} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

2961. $L_{263} \circ L_{342}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1423) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

2962. $L_{263} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

2963. $L_{263} \circ L_{384}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2964. $L_{263} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2965. $L_{263} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2966. $L_{263} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2967. $L_{263} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

2968. $L_{263} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2969. $L_{263} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2970. $L_{263} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2971. $L_{263} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

2972. $L_{263} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2973. $L_{263} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

2974. $L_{263} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2975. $L_{263} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \\ 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

2976. $L_{263} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2977. $L_{263} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2978. $L_{263} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

2979. $L_{263} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2980. $L_{264} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2981. $L_{264} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2982. $L_{264} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2983. $L_{264} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

2984. $L_{264} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2985. $L_{264} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2986. $L_{264} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2987. $L_{264} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

2988. $L_{264} \circ L_{511}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2989. $L_{264} \circ L_{520}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2990. $L_{264} \circ L_{541}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2991. $L_{264} \circ L_{552}$

$$A = \begin{pmatrix} 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1324) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

2992. $L_{265} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2993. $L_{265} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2994. $L_{265} \circ L_{320}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

2995. $L_{265} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2996. $L_{265} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

2997. $L_{265} \circ L_{376}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

2998. $L_{265} \circ L_{400}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 7 & 8 \\ 7 & 5 & 8 & 7 \\ 7 & 8 & 5 & 7 \\ 8 & 7 & 7 & 5 \end{pmatrix}$$

2999. $L_{265} \circ L_{409}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3000. $L_{265} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3001. $L_{265} \circ L_{464}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3002. $L_{265} \circ L_{503}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 7 \\ 5 & 7 & 7 & 8 \\ 8 & 7 & 7 & 5 \\ 7 & 5 & 8 & 7 \end{pmatrix}$$

3003. $L_{265} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3004. $L_{265} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3005. $L_{265} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3006. $L_{265} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3007. $L_{265} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3008. $L_{267} \circ L_{268}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3009. $L_{267} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3010. $L_{267} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3011. $L_{267} \circ L_{313}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3012. $L_{267} \circ L_{315}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3013. $L_{267} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3014. $L_{267} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3015. $L_{267} \circ L_{369}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3016. $L_{267} \circ L_{371}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3017. $L_{267} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3018. $L_{267} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3019. $L_{267} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3020. $L_{267} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3021. $L_{267} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3022. $L_{267} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3023. $L_{268} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3024. $L_{268} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3025. $L_{268} \circ L_{313}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3026. $L_{268} \circ L_{315}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3027. $L_{268} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3028. $L_{268} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3029. $L_{268} \circ L_{369}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (1)(234) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3030. $L_{268} \circ L_{371}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3031. $L_{268} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

3032. $L_{268} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3033. $L_{268} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3034. $L_{268} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3035. $L_{268} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3036. $L_{268} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3037. $L_{269} \circ L_{270}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(234) \\ \tau_3 = (143)(2) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3038. $L_{272} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3039. $L_{272} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

3040. $L_{272} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3041. $L_{272} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3042. $L_{273} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3043. $L_{273} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3044. $L_{273} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 5 \\ 6 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

3045. $L_{273} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3046. $L_{273} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3047. $L_{273} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3048. $L_{273} \circ L_{382}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3049. $L_{273} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3050. $L_{273} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3051. $L_{273} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3052. $L_{273} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3053. $L_{273} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3054. $L_{273} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3055. $L_{273} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3056. $L_{273} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3057. $L_{273} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3058. $L_{273} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 6 \\ 5 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3059. $L_{273} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1)(24)(3) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3060. $L_{273} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3061. $L_{273} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 5 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3062. $L_{273} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3063. $L_{273} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3064. $L_{273} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

3065. $L_{274} \circ L_{277}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3066. $L_{274} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3067. $L_{274} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3068. $L_{274} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3069. $L_{274} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3070. $L_{274} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3071. $L_{274} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3072. $L_{274} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3073. $L_{274} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3074. $L_{274} \circ L_{488}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3075. $L_{274} \circ L_{493}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3076. $L_{274} \circ L_{568}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3077. $L_{274} \circ L_{569}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(243) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3078. $L_{275} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3079. $L_{275} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3080. $L_{275} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3081. $L_{275} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3082. $L_{275} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3083. $L_{275} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3084. $L_{275} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3085. $L_{275} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3086. $L_{275} \circ L_{495}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3087. $L_{275} \circ L_{566}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3088. $L_{276} \circ L_{278}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3089. $L_{276} \circ L_{291}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3090. $L_{276} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3091. $L_{276} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3092. $L_{276} \circ L_{325}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3093. $L_{276} \circ L_{326}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3094. $L_{276} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3095. $L_{276} \circ L_{340}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

3096. $L_{276} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3097. $L_{276} \circ L_{381}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

3098. $L_{276} \circ L_{382}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3099. $L_{276} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3100. $L_{276} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3101. $L_{276} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3102. $L_{276} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3103. $L_{276} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3104. $L_{276} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3105. $L_{276} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3106. $L_{276} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3107. $L_{276} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3108. $L_{276} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \\ 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

3109. $L_{276} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3110. $L_{276} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3111. $L_{276} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 5 & 7 & 6 & 8 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

3112. $L_{276} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3113. $L_{276} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3114. $L_{276} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3115. $L_{276} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3116. $L_{276} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3117. $L_{277} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3118. $L_{277} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3119. $L_{277} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3120. $L_{277} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3121. $L_{277} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3122. $L_{277} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3123. $L_{277} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3124. $L_{277} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3125. $L_{277} \circ L_{488}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3126. $L_{277} \circ L_{493}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3127. $L_{277} \circ L_{568}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3128. $L_{277} \circ L_{569}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(34) \\ \tau_3 = (1423) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3129. $L_{278} \circ L_{289}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3130. $L_{278} \circ L_{290}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3131. $L_{278} \circ L_{327}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3132. $L_{278} \circ L_{328}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3133. $L_{278} \circ L_{341}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

3134. $L_{278} \circ L_{342}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3135. $L_{278} \circ L_{383}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3136. $L_{278} \circ L_{384}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

3137. $L_{278} \circ L_{405}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3138. $L_{278} \circ L_{407}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3139. $L_{278} \circ L_{420}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3140. $L_{278} \circ L_{422}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3141. $L_{278} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3142. $L_{278} \circ L_{436}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3143. $L_{278} \circ L_{469}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3144. $L_{278} \circ L_{471}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3145. $L_{278} \circ L_{494}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \\ 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

3146. $L_{278} \circ L_{496}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3147. $L_{278} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

3148. $L_{278} \circ L_{519}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3149. $L_{278} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3150. $L_{278} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3151. $L_{278} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3152. $L_{278} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3153. $L_{279} \circ L_{291}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3154. $L_{279} \circ L_{292}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3155. $L_{279} \circ L_{303}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3156. $L_{279} \circ L_{323}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3157. $L_{279} \circ L_{339}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3158. $L_{279} \circ L_{360}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3159. $L_{279} \circ L_{379}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 7 & 8 \\ 7 & 5 & 8 & 7 \\ 7 & 8 & 5 & 7 \\ 8 & 7 & 7 & 5 \end{pmatrix}$$

3160. $L_{279} \circ L_{434}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3161. $L_{279} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3162. $L_{279} \circ L_{517}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3163. $L_{279} \circ L_{542}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3164. $L_{279} \circ L_{544}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3165. $L_{279} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3166. $L_{279} \circ L_{565}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3167. $L_{279} \circ L_{567}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(23)(4) \\ \tau_3 &= (14)(2)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3168. $L_{281} \circ L_{283}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1)(2)(34) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$B = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (1)(243) \\ \tau_3 &= (142)(3) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3169. $L_{282} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3170. $L_{282} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 5 \end{pmatrix}$$

3171. $L_{282} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3172. $L_{282} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3173. $L_{282} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3174. $L_{282} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3175. $L_{282} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3176. $L_{282} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3177. $L_{282} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3178. $L_{282} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3179. $L_{282} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1)(2)(3)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3180. $L_{282} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1)(2)(3)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 5 \\ 5 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

3181. $L_{285} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (142)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

3182. $L_{285} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(243) \\ \tau_3 = (142)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3183. $L_{286} \circ L_{311}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3184. $L_{286} \circ L_{330}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3185. $L_{286} \circ L_{346}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3186. $L_{286} \circ L_{367}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \end{pmatrix}$$

3187. $L_{286} \circ L_{391}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3188. $L_{286} \circ L_{426}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3189. $L_{286} \circ L_{455}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3190. $L_{286} \circ L_{474}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3191. $L_{286} \circ L_{487}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

3192. $L_{286} \circ L_{522}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3193. $L_{286} \circ L_{535}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3194. $L_{286} \circ L_{570}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1)(24)(3) \\ \tau_3 = (1432) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3195. $L_{287} \circ L_{447}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3196. $L_{287} \circ L_{512}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 5 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3197. $L_{287} \circ L_{551}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 5 & 7 & 8 & 6 \end{pmatrix}$$

3198. $L_{287} \circ L_{560}$

$$A = \begin{pmatrix} 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (1)(23)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3199. $L_{289} \circ L_{304}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3200. $L_{289} \circ L_{342}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3201. $L_{289} \circ L_{359}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3202. $L_{289} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3203. $L_{289} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3204. $L_{289} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3205. $L_{289} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3206. $L_{289} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3207. $L_{289} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3208. $L_{289} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3209. $L_{289} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3210. $L_{289} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3211. $L_{289} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3212. $L_{290} \circ L_{327}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 7 \\ 5 & 8 & 7 & 7 \\ 7 & 7 & 8 & 5 \\ 7 & 7 & 5 & 8 \end{pmatrix}$$

3213. $L_{290} \circ L_{341}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3214. $L_{290} \circ L_{383}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3215. $L_{290} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3216. $L_{290} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3217. $L_{290} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3218. $L_{290} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3219. $L_{290} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3220. $L_{290} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3221. $L_{290} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3222. $L_{290} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3223. $L_{290} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3224. $L_{290} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3225. $L_{290} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 5 & 8 \\ 7 & 7 & 8 & 5 \\ 8 & 5 & 7 & 7 \\ 5 & 8 & 7 & 7 \end{pmatrix}$$

3226. $L_{290} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3227. $L_{291} \circ L_{340}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3228. $L_{291} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3229. $L_{291} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3230. $L_{291} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3231. $L_{291} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3232. $L_{291} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3233. $L_{291} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3234. $L_{291} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3235. $L_{291} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3236. $L_{291} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3237. $L_{291} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3238. $L_{291} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3239. $L_{291} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3240. $L_{291} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3241. $L_{292} \circ L_{303}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 5 & 8 \end{pmatrix}$$

3242. $L_{292} \circ L_{313}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3243. $L_{292} \circ L_{315}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3244. $L_{292} \circ L_{320}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3245. $L_{292} \circ L_{323}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3246. $L_{292} \circ L_{325}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3247. $L_{292} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3248. $L_{292} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

3249. $L_{292} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

3250. $L_{292} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3251. $L_{292} \circ L_{369}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3252. $L_{292} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3253. $L_{292} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3254. $L_{292} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3255. $L_{292} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3256. $L_{292} \circ L_{390}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3257. $L_{292} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3258. $L_{292} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3259. $L_{292} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3260. $L_{292} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3261. $L_{292} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3262. $L_{292} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3263. $L_{292} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3264. $L_{292} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3265. $L_{292} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3266. $L_{292} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3267. $L_{292} \circ L_{431}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3268. $L_{292} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3269. $L_{292} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3270. $L_{292} \circ L_{439}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3271. $L_{292} \circ L_{444}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3272. $L_{292} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3273. $L_{292} \circ L_{450}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3274. $L_{292} \circ L_{459}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3275. $L_{292} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3276. $L_{292} \circ L_{467}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3277. $L_{292} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3278. $L_{292} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3279. $L_{292} \circ L_{479}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3280. $L_{292} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3281. $L_{292} \circ L_{490}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3282. $L_{292} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3283. $L_{292} \circ L_{499}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3284. $L_{292} \circ L_{501}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3285. $L_{292} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3286. $L_{292} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 5 \\ 5 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3287. $L_{292} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3288. $L_{292} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 5 \\ 5 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

3289. $L_{292} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

3290. $L_{292} \circ L_{553}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3291. $L_{292} \circ L_{555}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3292. $L_{292} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3293. $L_{292} \circ L_{563}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3294. $L_{292} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3295. $L_{292} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3296. $L_{293} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3297. $L_{293} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3298. $L_{293} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3299. $L_{293} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (1423) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3300. $L_{294} \circ L_{323}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3301. $L_{294} \circ L_{325}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3302. $L_{294} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3303. $L_{294} \circ L_{337}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3304. $L_{294} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3305. $L_{294} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3306. $L_{294} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

3307. $L_{294} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

3308. $L_{294} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

3309. $L_{294} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3310. $L_{294} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 4 & 1 & 3 & 2 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3311. $L_{294} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 5 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3312. $L_{294} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3313. $L_{294} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 7 \\ 5 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3314. $L_{294} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 7 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3315. $L_{294} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3316. $L_{294} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3317. $L_{294} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3318. $L_{295} \circ L_{301}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3319. $L_{295} \circ L_{344}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3320. $L_{295} \circ L_{355}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3321. $L_{296} \circ L_{538}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3322. $L_{300} \circ L_{547}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3323. $L_{301} \circ L_{344}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3324. $L_{301} \circ L_{355}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3325. $L_{303} \circ L_{313}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3326. $L_{303} \circ L_{315}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3327. $L_{303} \circ L_{320}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3328. $L_{303} \circ L_{323}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3329. $L_{303} \circ L_{325}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3330. $L_{303} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3331. $L_{303} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 5 \\ 5 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

3332. $L_{303} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

3333. $L_{303} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3334. $L_{303} \circ L_{369}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3335. $L_{303} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3336. $L_{303} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3337. $L_{303} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3338. $L_{303} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3339. $L_{303} \circ L_{390}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3340. $L_{303} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3341. $L_{303} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3342. $L_{303} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3343. $L_{303} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3344. $L_{303} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3345. $L_{303} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3346. $L_{303} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3347. $L_{303} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3348. $L_{303} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3349. $L_{303} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3350. $L_{303} \circ L_{431}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3351. $L_{303} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3352. $L_{303} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3353. $L_{303} \circ L_{439}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3354. $L_{303} \circ L_{444}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3355. $L_{303} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3356. $L_{303} \circ L_{450}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3357. $L_{303} \circ L_{459}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3358. $L_{303} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3359. $L_{303} \circ L_{467}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3360. $L_{303} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3361. $L_{303} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3362. $L_{303} \circ L_{479}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3363. $L_{303} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3364. $L_{303} \circ L_{490}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3365. $L_{303} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3366. $L_{303} \circ L_{499}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3367. $L_{303} \circ L_{501}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3368. $L_{303} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3369. $L_{303} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3370. $L_{303} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3371. $L_{303} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

3372. $L_{303} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

3373. $L_{303} \circ L_{553}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3374. $L_{303} \circ L_{555}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3375. $L_{303} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3376. $L_{303} \circ L_{563}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3377. $L_{303} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3378. $L_{303} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3379. $L_{304} \circ L_{342}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3380. $L_{304} \circ L_{359}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3381. $L_{304} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3382. $L_{304} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3383. $L_{304} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3384. $L_{304} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3385. $L_{304} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3386. $L_{304} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3387. $L_{304} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3388. $L_{304} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3389. $L_{304} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3390. $L_{304} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

3391. $L_{305} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3392. $L_{305} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3393. $L_{307} \circ L_{309}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3394. $L_{307} \circ L_{325}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3395. $L_{307} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3396. $L_{307} \circ L_{348}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3397. $L_{307} \circ L_{350}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3398. $L_{307} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3399. $L_{307} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3400. $L_{307} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3401. $L_{307} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3402. $L_{307} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3403. $L_{307} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3404. $L_{307} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3405. $L_{307} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

3406. $L_{307} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3407. $L_{307} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3408. $L_{308} \circ L_{310}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(243) \\ \tau_4 = (142)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (142)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3409. $L_{308} \circ L_{347}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(243) \\ \tau_4 = (142)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(243) \\ \tau_4 = (142)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3410. $L_{308} \circ L_{349}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1)(243) \\ \tau_4 = (142)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (142)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3411. $L_{309} \circ L_{325}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3412. $L_{309} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3413. $L_{309} \circ L_{348}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3414. $L_{309} \circ L_{350}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3415. $L_{309} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3416. $L_{309} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3417. $L_{309} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3418. $L_{309} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3419. $L_{309} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3420. $L_{309} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3421. $L_{309} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3422. $L_{309} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3423. $L_{310} \circ L_{347}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3424. $L_{310} \circ L_{349}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3425. $L_{311} \circ L_{325}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3426. $L_{311} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3427. $L_{311} \circ L_{346}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3428. $L_{311} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3429. $L_{311} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3430. $L_{311} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3431. $L_{311} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3432. $L_{311} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

3433. $L_{311} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 5 & 8 \\ 8 & 7 & 6 & 5 \\ 5 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

3434. $L_{311} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3435. $L_{311} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3436. $L_{311} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3437. $L_{311} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3438. $L_{311} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3439. $L_{311} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3440. $L_{313} \circ L_{315}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3441. $L_{313} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3442. $L_{313} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3443. $L_{313} \circ L_{369}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3444. $L_{313} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3445. $L_{313} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3446. $L_{313} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3447. $L_{313} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3448. $L_{313} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3449. $L_{313} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3450. $L_{313} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3451. $L_{314} \circ L_{316}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3452. $L_{314} \circ L_{370}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3453. $L_{314} \circ L_{372}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3454. $L_{315} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3455. $L_{315} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3456. $L_{315} \circ L_{369}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3457. $L_{315} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3458. $L_{315} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3459. $L_{315} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3460. $L_{315} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3461. $L_{315} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3462. $L_{315} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3463. $L_{315} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3464. $L_{315} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3465. $L_{315} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3466. $L_{316} \circ L_{370}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3467. $L_{316} \circ L_{372}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3468. $L_{318} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3469. $L_{318} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3470. $L_{320} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3471. $L_{320} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3472. $L_{320} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 6 & 7 & 8 \\ 7 & 6 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

3473. $L_{320} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3474. $L_{320} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3475. $L_{320} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 8 & 7 & 5 & 6 \\ 6 & 5 & 7 & 8 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3476. $L_{320} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3477. $L_{320} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3478. $L_{320} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3479. $L_{320} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3480. $L_{320} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3481. $L_{320} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3482. $L_{320} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3483. $L_{321} \circ L_{561}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3484. $L_{322} \circ L_{331}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3485. $L_{322} \circ L_{366}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3486. $L_{322} \circ L_{378}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(243) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3487. $L_{323} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3488. $L_{323} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3489. $L_{323} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3490. $L_{323} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3491. $L_{323} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3492. $L_{323} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3493. $L_{323} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3494. $L_{323} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3495. $L_{323} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3496. $L_{323} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3497. $L_{323} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3498. $L_{323} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3499. $L_{323} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3500. $L_{323} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3501. $L_{323} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3502. $L_{324} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 5 & 6 \end{pmatrix}$$

3503. $L_{324} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

3504. $L_{324} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3505. $L_{324} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3506. $L_{325} \circ L_{330}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

3507. $L_{325} \circ L_{337}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3508. $L_{325} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3509. $L_{325} \circ L_{346}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3510. $L_{325} \circ L_{348}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3511. $L_{325} \circ L_{350}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3512. $L_{325} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3513. $L_{325} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 5 & 8 \end{pmatrix}$$

3514. $L_{325} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

3515. $L_{325} \circ L_{386}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3516. $L_{325} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3517. $L_{325} \circ L_{393}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3518. $L_{325} \circ L_{401}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3519. $L_{325} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3520. $L_{325} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3521. $L_{325} \circ L_{416}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3522. $L_{325} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3523. $L_{325} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3524. $L_{325} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3525. $L_{325} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3526. $L_{325} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3527. $L_{325} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3528. $L_{325} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3529. $L_{325} \circ L_{437}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3530. $L_{325} \circ L_{442}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3531. $L_{325} \circ L_{451}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3532. $L_{325} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3533. $L_{325} \circ L_{461}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3534. $L_{325} \circ L_{465}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3535. $L_{325} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3536. $L_{325} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3537. $L_{325} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3538. $L_{325} \circ L_{478}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3539. $L_{325} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3540. $L_{325} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3541. $L_{325} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3542. $L_{325} \circ L_{513}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3543. $L_{325} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3544. $L_{325} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3545. $L_{325} \circ L_{524}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3546. $L_{325} \circ L_{526}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3547. $L_{325} \circ L_{530}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3548. $L_{325} \circ L_{532}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3549. $L_{325} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3550. $L_{325} \circ L_{540}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3551. $L_{325} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3552. $L_{325} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3553. $L_{325} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 6 & 7 & 5 & 8 \\ 8 & 5 & 7 & 6 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3554. $L_{325} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3555. $L_{326} \circ L_{382}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3556. $L_{326} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3557. $L_{326} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3558. $L_{326} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3559. $L_{326} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3560. $L_{326} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3561. $L_{326} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3562. $L_{326} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3563. $L_{326} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3564. $L_{326} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3565. $L_{326} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3566. $L_{326} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3567. $L_{326} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3568. $L_{326} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3569. $L_{327} \circ L_{341}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3570. $L_{327} \circ L_{383}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 6 \end{pmatrix}$$

3571. $L_{327} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3572. $L_{327} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3573. $L_{327} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3574. $L_{327} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3575. $L_{327} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3576. $L_{327} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3577. $L_{327} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3578. $L_{327} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3579. $L_{327} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3580. $L_{327} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3581. $L_{327} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3582. $L_{327} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 5 \\ 7 & 7 & 5 & 8 \\ 5 & 8 & 7 & 7 \\ 8 & 5 & 7 & 7 \end{pmatrix}$$

3583. $L_{328} \circ L_{329}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3584. $L_{328} \circ L_{368}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3585. $L_{328} \circ L_{384}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3586. $L_{328} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3587. $L_{328} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3588. $L_{328} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3589. $L_{328} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3590. $L_{328} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3591. $L_{328} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3592. $L_{328} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3593. $L_{328} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3594. $L_{328} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3595. $L_{328} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3596. $L_{329} \circ L_{368}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3597. $L_{329} \circ L_{384}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3598. $L_{329} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3599. $L_{329} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3600. $L_{329} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3601. $L_{329} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3602. $L_{329} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3603. $L_{329} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3604. $L_{329} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3605. $L_{329} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3606. $L_{329} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3607. $L_{329} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3608. $L_{330} \circ L_{337}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3609. $L_{330} \circ L_{339}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3610. $L_{330} \circ L_{346}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3611. $L_{330} \circ L_{348}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3612. $L_{330} \circ L_{350}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3613. $L_{330} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3614. $L_{330} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

3615. $L_{330} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \end{pmatrix}$$

3616. $L_{330} \circ L_{386}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3617. $L_{330} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3618. $L_{330} \circ L_{393}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3619. $L_{330} \circ L_{401}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3620. $L_{330} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3621. $L_{330} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3622. $L_{330} \circ L_{416}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3623. $L_{330} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3624. $L_{330} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3625. $L_{330} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3626. $L_{330} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3627. $L_{330} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3628. $L_{330} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3629. $L_{330} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3630. $L_{330} \circ L_{437}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3631. $L_{330} \circ L_{442}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3632. $L_{330} \circ L_{451}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3633. $L_{330} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3634. $L_{330} \circ L_{461}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3635. $L_{330} \circ L_{465}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3636. $L_{330} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3637. $L_{330} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3638. $L_{330} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3639. $L_{330} \circ L_{478}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3640. $L_{330} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3641. $L_{330} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

3642. $L_{330} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3643. $L_{330} \circ L_{513}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3644. $L_{330} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3645. $L_{330} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3646. $L_{330} \circ L_{524}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3647. $L_{330} \circ L_{526}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3648. $L_{330} \circ L_{530}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3649. $L_{330} \circ L_{532}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3650. $L_{330} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3651. $L_{330} \circ L_{540}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3652. $L_{330} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3653. $L_{330} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3654. $L_{330} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3655. $L_{330} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 5 \\ 5 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

3656. $L_{331} \circ L_{366}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3657. $L_{331} \circ L_{378}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (134)(2) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3658. $L_{334} \circ L_{574}$

$$A = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (13)(2)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(2)(4) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3659. $L_{337} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3660. $L_{337} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3661. $L_{337} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3662. $L_{337} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

3663. $L_{337} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

3664. $L_{337} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3665. $L_{337} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3666. $L_{337} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3667. $L_{337} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 6 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3668. $L_{337} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3669. $L_{337} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3670. $L_{338} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(243) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3671. $L_{338} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(243) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3672. $L_{339} \circ L_{360}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \end{pmatrix}$$

3673. $L_{339} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3674. $L_{339} \circ L_{369}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3675. $L_{339} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3676. $L_{339} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1342) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3677. $L_{339} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3678. $L_{339} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3679. $L_{339} \circ L_{390}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3680. $L_{339} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3681. $L_{339} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3682. $L_{339} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3683. $L_{339} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3684. $L_{339} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3685. $L_{339} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3686. $L_{339} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3687. $L_{339} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3688. $L_{339} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3689. $L_{339} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3690. $L_{339} \circ L_{431}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3691. $L_{339} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3692. $L_{339} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3693. $L_{339} \circ L_{439}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3694. $L_{339} \circ L_{444}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3695. $L_{339} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3696. $L_{339} \circ L_{450}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3697. $L_{339} \circ L_{459}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3698. $L_{339} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3699. $L_{339} \circ L_{467}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3700. $L_{339} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3701. $L_{339} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3702. $L_{339} \circ L_{479}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3703. $L_{339} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3704. $L_{339} \circ L_{490}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3705. $L_{339} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3706. $L_{339} \circ L_{499}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3707. $L_{339} \circ L_{501}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3708. $L_{339} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3709. $L_{339} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

3710. $L_{339} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

3711. $L_{339} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3712. $L_{339} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3713. $L_{339} \circ L_{553}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3714. $L_{339} \circ L_{555}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3715. $L_{339} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3716. $L_{339} \circ L_{563}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3717. $L_{339} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3718. $L_{339} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3719. $L_{340} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3720. $L_{340} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3721. $L_{340} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3722. $L_{340} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3723. $L_{340} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3724. $L_{340} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3725. $L_{340} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3726. $L_{340} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3727. $L_{340} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3728. $L_{340} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3729. $L_{340} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

3730. $L_{340} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3731. $L_{340} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3732. $L_{340} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3733. $L_{340} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3734. $L_{340} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (14)(23) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3735. $L_{340} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

3736. $L_{341} \circ L_{383}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3737. $L_{341} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3738. $L_{341} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3739. $L_{341} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3740. $L_{341} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3741. $L_{341} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3742. $L_{341} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3743. $L_{341} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3744. $L_{341} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3745. $L_{341} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

3746. $L_{341} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3747. $L_{341} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3748. $L_{341} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3749. $L_{342} \circ L_{359}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3750. $L_{342} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3751. $L_{342} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3752. $L_{342} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3753. $L_{342} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3754. $L_{342} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3755. $L_{342} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3756. $L_{342} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3757. $L_{342} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3758. $L_{342} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 5 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3759. $L_{342} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

3760. $L_{342} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3761. $L_{342} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3762. $L_{343} \circ L_{515}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3763. $L_{344} \circ L_{355}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (143)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3764. $L_{346} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3765. $L_{346} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3766. $L_{346} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

3767. $L_{346} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \end{pmatrix}$$

3768. $L_{346} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

3769. $L_{346} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

3770. $L_{346} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3771. $L_{346} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3772. $L_{346} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \end{pmatrix}$$

3773. $L_{346} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3774. $L_{346} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3775. $L_{346} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3776. $L_{347} \circ L_{349}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (142)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3777. $L_{348} \circ L_{350}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3778. $L_{348} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3779. $L_{348} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3780. $L_{348} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3781. $L_{348} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3782. $L_{348} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3783. $L_{348} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

3784. $L_{348} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3785. $L_{348} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3786. $L_{350} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3787. $L_{350} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3788. $L_{350} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3789. $L_{350} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3790. $L_{350} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3791. $L_{350} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3792. $L_{350} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3793. $L_{350} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3794. $L_{352} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (1432) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

3795. $L_{352} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3796. $L_{352} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

3797. $L_{352} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (1432) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3798. $L_{358} \circ L_{507}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (14)(2)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(2)(3) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3799. $L_{359} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3800. $L_{359} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3801. $L_{359} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3802. $L_{359} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3803. $L_{359} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3804. $L_{359} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3805. $L_{359} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3806. $L_{359} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3807. $L_{359} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3808. $L_{359} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1423) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3809. $L_{360} \circ L_{367}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3810. $L_{360} \circ L_{369}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3811. $L_{360} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3812. $L_{360} \circ L_{376}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3813. $L_{360} \circ L_{379}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3814. $L_{360} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3815. $L_{360} \circ L_{390}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3816. $L_{360} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3817. $L_{360} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3818. $L_{360} \circ L_{403}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3819. $L_{360} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3820. $L_{360} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3821. $L_{360} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3822. $L_{360} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3823. $L_{360} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3824. $L_{360} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3825. $L_{360} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3826. $L_{360} \circ L_{431}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3827. $L_{360} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3828. $L_{360} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3829. $L_{360} \circ L_{439}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3830. $L_{360} \circ L_{444}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3831. $L_{360} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3832. $L_{360} \circ L_{450}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3833. $L_{360} \circ L_{459}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3834. $L_{360} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3835. $L_{360} \circ L_{467}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3836. $L_{360} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3837. $L_{360} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3838. $L_{360} \circ L_{479}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3839. $L_{360} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3840. $L_{360} \circ L_{490}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3841. $L_{360} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3842. $L_{360} \circ L_{499}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3843. $L_{360} \circ L_{501}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3844. $L_{360} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3845. $L_{360} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 6 & 7 & 8 & 5 \\ 5 & 8 & 7 & 6 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

3846. $L_{360} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

3847. $L_{360} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 5 \\ 5 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3848. $L_{360} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 6 & 8 \\ 6 & 7 & 8 & 6 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3849. $L_{360} \circ L_{553}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3850. $L_{360} \circ L_{555}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3851. $L_{360} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3852. $L_{360} \circ L_{563}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3853. $L_{360} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3854. $L_{360} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (14)(23) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3855. $L_{363} \circ L_{482}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3856. $L_{366} \circ L_{378}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (134)(2) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(243) \\ \tau_4 = (134)(2) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3857. $L_{367} \circ L_{381}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

3858. $L_{367} \circ L_{386}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3859. $L_{367} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3860. $L_{367} \circ L_{393}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3861. $L_{367} \circ L_{401}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3862. $L_{367} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3863. $L_{367} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3864. $L_{367} \circ L_{416}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3865. $L_{367} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3866. $L_{367} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3867. $L_{367} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3868. $L_{367} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3869. $L_{367} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3870. $L_{367} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3871. $L_{367} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3872. $L_{367} \circ L_{437}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3873. $L_{367} \circ L_{442}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3874. $L_{367} \circ L_{451}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3875. $L_{367} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3876. $L_{367} \circ L_{461}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3877. $L_{367} \circ L_{465}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3878. $L_{367} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3879. $L_{367} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3880. $L_{367} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3881. $L_{367} \circ L_{478}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3882. $L_{367} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 5 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3883. $L_{367} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3884. $L_{367} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3885. $L_{367} \circ L_{513}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3886. $L_{367} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3887. $L_{367} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3888. $L_{367} \circ L_{524}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3889. $L_{367} \circ L_{526}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3890. $L_{367} \circ L_{530}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3891. $L_{367} \circ L_{532}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3892. $L_{367} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3893. $L_{367} \circ L_{540}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3894. $L_{367} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3895. $L_{367} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3896. $L_{367} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3897. $L_{367} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3898. $L_{368} \circ L_{384}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3899. $L_{368} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3900. $L_{368} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3901. $L_{368} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3902. $L_{368} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3903. $L_{368} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3904. $L_{368} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3905. $L_{368} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3906. $L_{368} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3907. $L_{368} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

3908. $L_{368} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3909. $L_{369} \circ L_{371}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3910. $L_{369} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 5 & 8 & 6 & 7 \end{pmatrix}$$

3911. $L_{369} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3912. $L_{369} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3913. $L_{369} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3914. $L_{369} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3915. $L_{369} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3916. $L_{370} \circ L_{372}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (132)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3917. $L_{371} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3918. $L_{371} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3919. $L_{371} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3920. $L_{371} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3921. $L_{371} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3922. $L_{371} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3923. $L_{374} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3924. $L_{374} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 5 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3925. $L_{374} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

3926. $L_{374} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3927. $L_{376} \circ L_{400}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 5 \\ 7 & 8 & 5 & 7 \\ 7 & 5 & 8 & 7 \\ 5 & 7 & 7 & 8 \end{pmatrix}$$

3928. $L_{376} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3929. $L_{376} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3930. $L_{376} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3931. $L_{376} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 7 \\ 8 & 7 & 7 & 5 \\ 5 & 7 & 7 & 8 \\ 7 & 8 & 5 & 7 \end{pmatrix}$$

3932. $L_{376} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3933. $L_{376} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3934. $L_{376} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3935. $L_{376} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3936. $L_{376} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1342) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3937. $L_{377} \circ L_{492}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (13)(2)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(2)(4) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3938. $L_{379} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3939. $L_{379} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3940. $L_{379} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3941. $L_{379} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3942. $L_{379} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3943. $L_{379} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3944. $L_{379} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3945. $L_{379} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3946. $L_{380} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(234) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3947. $L_{380} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(234) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3948. $L_{381} \circ L_{386}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3949. $L_{381} \circ L_{391}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3950. $L_{381} \circ L_{393}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3951. $L_{381} \circ L_{401}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3952. $L_{381} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3953. $L_{381} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3954. $L_{381} \circ L_{416}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3955. $L_{381} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

3956. $L_{381} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3957. $L_{381} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

3958. $L_{381} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

3959. $L_{381} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

3960. $L_{381} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3961. $L_{381} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3962. $L_{381} \circ L_{437}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3963. $L_{381} \circ L_{442}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3964. $L_{381} \circ L_{451}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3965. $L_{381} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1432) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3966. $L_{381} \circ L_{461}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3967. $L_{381} \circ L_{465}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3968. $L_{381} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3969. $L_{381} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3970. $L_{381} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3971. $L_{381} \circ L_{478}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3972. $L_{381} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3973. $L_{381} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

3974. $L_{381} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3975. $L_{381} \circ L_{513}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3976. $L_{381} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3977. $L_{381} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3978. $L_{381} \circ L_{524}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3979. $L_{381} \circ L_{526}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3980. $L_{381} \circ L_{530}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3981. $L_{381} \circ L_{532}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

3982. $L_{381} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

3983. $L_{381} \circ L_{540}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

3984. $L_{381} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3985. $L_{381} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3986. $L_{381} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \end{pmatrix}$$

3987. $L_{381} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \end{pmatrix}$$

3988. $L_{382} \circ L_{401}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

3989. $L_{382} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (13)(24) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3990. $L_{382} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3991. $L_{382} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3992. $L_{382} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

3993. $L_{382} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3994. $L_{382} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3995. $L_{382} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3996. $L_{382} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

3997. $L_{382} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

3998. $L_{382} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

3999. $L_{382} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4000. $L_{382} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \end{pmatrix}$$

4001. $L_{382} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4002. $L_{382} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \end{pmatrix}$$

4003. $L_{382} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4004. $L_{382} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (13)(24) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4005. $L_{383} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4006. $L_{383} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4007. $L_{383} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4008. $L_{383} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4009. $L_{383} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4010. $L_{383} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4011. $L_{383} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4012. $L_{383} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4013. $L_{383} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4014. $L_{383} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 6 & 8 & 7 & 5 \\ 8 & 6 & 5 & 7 \end{pmatrix}$$

4015. $L_{383} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

4016. $L_{383} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4017. $L_{384} \circ L_{405}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4018. $L_{384} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4019. $L_{384} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (1324) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4020. $L_{384} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4021. $L_{384} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4022. $L_{384} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4023. $L_{384} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4024. $L_{384} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4025. $L_{384} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 5 & 7 \\ 8 & 6 & 7 & 5 \end{pmatrix}$$

4026. $L_{384} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4027. $L_{384} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4028. $L_{384} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

4029. $L_{385} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4030. $L_{385} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 4 & 2 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (1324) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4031. $L_{386} \circ L_{393}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4032. $L_{386} \circ L_{416}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4033. $L_{386} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4034. $L_{386} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

4035. $L_{386} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

4036. $L_{386} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4037. $L_{386} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4038. $L_{386} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4039. $L_{386} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4040. $L_{386} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4041. $L_{386} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4042. $L_{387} \circ L_{394}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4043. $L_{387} \circ L_{415}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4044. $L_{387} \circ L_{427}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4045. $L_{389} \circ L_{395}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4046. $L_{389} \circ L_{414}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4047. $L_{389} \circ L_{432}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4048. $L_{390} \circ L_{398}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4049. $L_{390} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4050. $L_{390} \circ L_{431}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4051. $L_{390} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4052. $L_{390} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4053. $L_{390} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4054. $L_{390} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4055. $L_{390} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4056. $L_{390} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4057. $L_{391} \circ L_{426}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \end{pmatrix}$$

4058. $L_{391} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4059. $L_{391} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4060. $L_{391} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4061. $L_{391} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4062. $L_{391} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

4063. $L_{391} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 5 \\ 5 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \end{pmatrix}$$

4064. $L_{391} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4065. $L_{391} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4066. $L_{393} \circ L_{416}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4067. $L_{393} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4068. $L_{393} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 5 \end{pmatrix}$$

4069. $L_{393} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4070. $L_{393} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4071. $L_{393} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4072. $L_{393} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4073. $L_{393} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4074. $L_{394} \circ L_{415}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4075. $L_{394} \circ L_{427}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(243) \\ \tau_4 = (123)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4076. $L_{395} \circ L_{414}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4077. $L_{395} \circ L_{432}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(234) \\ \tau_4 = (124)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4078. $L_{397} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4079. $L_{397} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

4080. $L_{397} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 5 & 6 & 7 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

4081. $L_{397} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(234) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4082. $L_{398} \circ L_{411}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4083. $L_{398} \circ L_{431}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4084. $L_{398} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4085. $L_{398} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4086. $L_{398} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4087. $L_{398} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4088. $L_{398} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4089. $L_{398} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4090. $L_{400} \circ L_{409}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 6 & 8 \\ 6 & 6 & 8 & 7 \\ 7 & 8 & 6 & 6 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

4091. $L_{400} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4092. $L_{400} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 7 \\ 5 & 7 & 7 & 8 \\ 8 & 7 & 7 & 5 \\ 7 & 5 & 8 & 7 \end{pmatrix}$$

4093. $L_{400} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4094. $L_{400} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4095. $L_{400} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4096. $L_{400} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4097. $L_{400} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4098. $L_{400} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \\ 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4099. $L_{401} \circ L_{417}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4100. $L_{401} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4101. $L_{401} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \\ 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4102. $L_{401} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 7 \\ 5 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4103. $L_{401} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4104. $L_{401} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4105. $L_{401} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

4106. $L_{401} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4107. $L_{401} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

4108. $L_{401} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4109. $L_{401} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 5 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 5 \end{pmatrix}$$

4110. $L_{401} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4111. $L_{401} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4112. $L_{403} \circ L_{423}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4113. $L_{403} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4114. $L_{403} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4115. $L_{403} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4116. $L_{403} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4117. $L_{403} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4118. $L_{403} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4119. $L_{403} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4120. $L_{403} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4121. $L_{405} \circ L_{407}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4122. $L_{405} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4123. $L_{405} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \end{pmatrix}$$

4124. $L_{405} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4125. $L_{405} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

4126. $L_{405} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \\ 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

4127. $L_{405} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4128. $L_{405} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4129. $L_{405} \circ L_{488}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4130. $L_{405} \circ L_{493}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4131. $L_{405} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4132. $L_{405} \circ L_{495}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4133. $L_{405} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4134. $L_{405} \circ L_{511}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4135. $L_{405} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4136. $L_{405} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4137. $L_{405} \circ L_{518}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4138. $L_{405} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4139. $L_{405} \circ L_{520}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4140. $L_{405} \circ L_{541}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4141. $L_{405} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4142. $L_{405} \circ L_{543}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4143. $L_{405} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4144. $L_{405} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4145. $L_{405} \circ L_{552}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4146. $L_{405} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4147. $L_{405} \circ L_{566}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4148. $L_{405} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4149. $L_{405} \circ L_{568}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4150. $L_{405} \circ L_{569}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4151. $L_{405} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4152. $L_{406} \circ L_{470}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4153. $L_{407} \circ L_{420}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \end{pmatrix}$$

4154. $L_{407} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4155. $L_{407} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4156. $L_{407} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4157. $L_{407} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4158. $L_{407} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 5 & 6 \\ 8 & 7 & 6 & 5 \\ 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4159. $L_{407} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4160. $L_{407} \circ L_{488}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4161. $L_{407} \circ L_{493}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4162. $L_{407} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4163. $L_{407} \circ L_{495}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4164. $L_{407} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4165. $L_{407} \circ L_{511}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4166. $L_{407} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4167. $L_{407} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4168. $L_{407} \circ L_{518}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4169. $L_{407} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4170. $L_{407} \circ L_{520}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4171. $L_{407} \circ L_{541}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4172. $L_{407} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4173. $L_{407} \circ L_{543}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4174. $L_{407} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4175. $L_{407} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4176. $L_{407} \circ L_{552}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4177. $L_{407} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4178. $L_{407} \circ L_{566}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4179. $L_{407} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4180. $L_{407} \circ L_{568}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4181. $L_{407} \circ L_{569}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4182. $L_{407} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4183. $L_{408} \circ L_{472}$

$$A = \begin{pmatrix} 3 & 4 & 1 & 2 \\ 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (1)(2)(34) \\ \tau_4 &= (12)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(3)(4) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4184. $L_{409} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4185. $L_{409} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4186. $L_{409} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 6 & 8 \\ 8 & 6 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4187. $L_{409} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4188. $L_{409} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4189. $L_{409} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4190. $L_{409} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4191. $L_{409} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 6 & 7 & 5 & 8 \\ 8 & 5 & 7 & 6 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4192. $L_{411} \circ L_{431}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4193. $L_{411} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 5 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4194. $L_{411} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4195. $L_{411} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4196. $L_{411} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4197. $L_{411} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4198. $L_{411} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4199. $L_{411} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4200. $L_{411} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4201. $L_{412} \circ L_{464}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1342) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4202. $L_{412} \circ L_{503}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (1243) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4203. $L_{414} \circ L_{432}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(234) \\ \tau_4 &= (124)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4204. $L_{415} \circ L_{427}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1342) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1)(243) \\ \tau_4 &= (123)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4205. $L_{416} \circ L_{430}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4206. $L_{416} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

4207. $L_{416} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4208. $L_{416} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4209. $L_{416} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4210. $L_{416} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4211. $L_{416} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (1234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4212. $L_{417} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4213. $L_{417} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4214. $L_{417} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1)(24)(3) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4215. $L_{417} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4216. $L_{417} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

4217. $L_{417} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4218. $L_{417} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 6 \end{pmatrix}$$

4219. $L_{417} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4220. $L_{419} \circ L_{433}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4221. $L_{420} \circ L_{422}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4222. $L_{420} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \\ 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4223. $L_{420} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4224. $L_{420} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4225. $L_{420} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 5 & 7 & 8 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4226. $L_{420} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4227. $L_{420} \circ L_{488}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4228. $L_{420} \circ L_{493}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4229. $L_{420} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4230. $L_{420} \circ L_{495}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4231. $L_{420} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4232. $L_{420} \circ L_{511}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4233. $L_{420} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4234. $L_{420} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4235. $L_{420} \circ L_{518}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4236. $L_{420} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4237. $L_{420} \circ L_{520}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4238. $L_{420} \circ L_{541}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4239. $L_{420} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4240. $L_{420} \circ L_{543}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4241. $L_{420} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4242. $L_{420} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4243. $L_{420} \circ L_{552}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4244. $L_{420} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4245. $L_{420} \circ L_{566}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4246. $L_{420} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4247. $L_{420} \circ L_{568}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4248. $L_{420} \circ L_{569}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4249. $L_{420} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4250. $L_{421} \circ L_{435}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(34) \\ \tau_4 = (12)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(3)(4) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4251. $L_{422} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4252. $L_{422} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 8 & 7 & 5 & 6 \\ 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

4253. $L_{422} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 5 & 6 & 7 & 8 \\ 6 & 5 & 8 & 7 \end{pmatrix}$$

4254. $L_{422} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 6 & 6 \\ 6 & 6 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4255. $L_{422} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4256. $L_{422} \circ L_{488}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4257. $L_{422} \circ L_{493}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4258. $L_{422} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4259. $L_{422} \circ L_{495}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4260. $L_{422} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4261. $L_{422} \circ L_{511}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4262. $L_{422} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4263. $L_{422} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4264. $L_{422} \circ L_{518}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4265. $L_{422} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4266. $L_{422} \circ L_{520}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4267. $L_{422} \circ L_{541}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4268. $L_{422} \circ L_{542}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4269. $L_{422} \circ L_{543}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4270. $L_{422} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4271. $L_{422} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4272. $L_{422} \circ L_{552}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (132)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4273. $L_{422} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4274. $L_{422} \circ L_{566}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4275. $L_{422} \circ L_{567}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4276. $L_{422} \circ L_{568}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4277. $L_{422} \circ L_{569}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4278. $L_{422} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (1)(2)(3)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4279. $L_{423} \circ L_{434}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4280. $L_{423} \circ L_{436}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (1423) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4281. $L_{423} \circ L_{469}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4282. $L_{423} \circ L_{471}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 5 & 6 & 8 & 7 \end{pmatrix}$$

4283. $L_{423} \circ L_{494}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4284. $L_{423} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 6 & 7 & 6 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4285. $L_{423} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4286. $L_{423} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4287. $L_{423} \circ L_{519}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 6 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4288. $L_{423} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4289. $L_{423} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1)(23)(4) \\ \tau_4 &= (12)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4290. $L_{423} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (12)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 8 & 5 & 7 & 6 \\ 6 & 7 & 5 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4291. $L_{426} \circ L_{455}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \\ 8 & 7 & 5 & 7 \end{pmatrix}$$

4292. $L_{426} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4293. $L_{426} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4294. $L_{426} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4295. $L_{426} \circ L_{522}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

4296. $L_{426} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \\ 8 & 6 & 6 & 7 \end{pmatrix}$$

4297. $L_{426} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4298. $L_{426} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1)(2)(3)(4) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4299. $L_{429} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4300. $L_{429} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 5 & 8 \\ 8 & 6 & 7 & 6 \end{pmatrix}$$

4301. $L_{429} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 6 & 7 & 7 \end{pmatrix}$$

4302. $L_{429} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(243) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \\ 8 & 5 & 7 & 6 \end{pmatrix}$$

4303. $L_{430} \circ L_{474}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 6 \end{pmatrix}$$

4304. $L_{430} \circ L_{487}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4305. $L_{430} \circ L_{496}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4306. $L_{430} \circ L_{535}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \end{pmatrix}$$

4307. $L_{430} \circ L_{565}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4308. $L_{430} \circ L_{570}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1)(24)(3) \\ \tau_4 = (1234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \end{pmatrix}$$

4309. $L_{431} \circ L_{447}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 7 \\ 6 & 6 & 7 & 8 \\ 8 & 7 & 5 & 6 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4310. $L_{431} \circ L_{512}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4311. $L_{431} \circ L_{517}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4312. $L_{431} \circ L_{544}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4313. $L_{431} \circ L_{551}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4314. $L_{431} \circ L_{560}$

$$A = \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1)(23)(4) \\ \tau_4 = (1243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 7 & 8 & 6 & 5 \\ 6 & 7 & 7 & 8 \\ 8 & 7 & 7 & 6 \\ 6 & 6 & 8 & 7 \end{pmatrix}$$

4315. $L_{434} \circ L_{436}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 5 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4316. $L_{434} \circ L_{447}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \end{pmatrix}$$

4317. $L_{434} \circ L_{464}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4318. $L_{434} \circ L_{469}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4319. $L_{434} \circ L_{471}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \end{pmatrix}$$

4320. $L_{434} \circ L_{474}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4321. $L_{434} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4322. $L_{434} \circ L_{488}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4323. $L_{434} \circ L_{493}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4324. $L_{434} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4325. $L_{434} \circ L_{495}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4326. $L_{434} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4327. $L_{434} \circ L_{503}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4328. $L_{434} \circ L_{511}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4329. $L_{434} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4330. $L_{434} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4331. $L_{434} \circ L_{518}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4332. $L_{434} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4333. $L_{434} \circ L_{520}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4334. $L_{434} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4335. $L_{434} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4336. $L_{434} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4337. $L_{434} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4338. $L_{434} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4339. $L_{434} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4340. $L_{434} \circ L_{552}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4341. $L_{434} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 5 \\ 5 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4342. $L_{434} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4343. $L_{434} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4344. $L_{434} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4345. $L_{434} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4346. $L_{434} \circ L_{569}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4347. $L_{434} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4348. $L_{436} \circ L_{447}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4349. $L_{436} \circ L_{469}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 5 & 8 & 7 \\ 5 & 6 & 7 & 8 \end{pmatrix}$$

4350. $L_{436} \circ L_{471}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4351. $L_{436} \circ L_{474}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4352. $L_{436} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4353. $L_{436} \circ L_{488}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4354. $L_{436} \circ L_{493}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4355. $L_{436} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4356. $L_{436} \circ L_{495}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4357. $L_{436} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4358. $L_{436} \circ L_{511}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4359. $L_{436} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4360. $L_{436} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4361. $L_{436} \circ L_{518}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4362. $L_{436} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4363. $L_{436} \circ L_{520}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4364. $L_{436} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \end{pmatrix}$$

4365. $L_{436} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4366. $L_{436} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4367. $L_{436} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4368. $L_{436} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4369. $L_{436} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4370. $L_{436} \circ L_{552}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4371. $L_{436} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 5 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4372. $L_{436} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4373. $L_{436} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4374. $L_{436} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4375. $L_{436} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4376. $L_{436} \circ L_{569}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4377. $L_{436} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (1423) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4378. $L_{437} \circ L_{465}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4379. $L_{437} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4380. $L_{437} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4381. $L_{437} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4382. $L_{437} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 1 & 2 & 4 \\ 1 & 2 & 4 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4383. $L_{439} \circ L_{467}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4384. $L_{439} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4385. $L_{439} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4386. $L_{439} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4387. $L_{439} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4388. $L_{441} \circ L_{454}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4389. $L_{441} \circ L_{462}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4390. $L_{441} \circ L_{475}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4391. $L_{442} \circ L_{451}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4392. $L_{442} \circ L_{461}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4393. $L_{442} \circ L_{478}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4394. $L_{442} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4395. $L_{442} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4396. $L_{442} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4397. $L_{442} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4398. $L_{444} \circ L_{450}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4399. $L_{444} \circ L_{459}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4400. $L_{444} \circ L_{479}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4401. $L_{444} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4402. $L_{444} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4403. $L_{444} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4404. $L_{444} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4405. $L_{445} \circ L_{449}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4406. $L_{445} \circ L_{458}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4407. $L_{445} \circ L_{480}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (143)(2) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4408. $L_{447} \circ L_{464}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4409. $L_{447} \circ L_{469}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (14)(23) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4410. $L_{447} \circ L_{471}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (1324) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 5 & 7 & 8 \end{pmatrix}$$

4411. $L_{447} \circ L_{474}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4412. $L_{447} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4413. $L_{447} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4414. $L_{447} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4415. $L_{447} \circ L_{503}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4416. $L_{447} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4417. $L_{447} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4418. $L_{447} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4419. $L_{447} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4420. $L_{447} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4421. $L_{447} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4422. $L_{447} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4423. $L_{447} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4424. $L_{447} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4425. $L_{449} \circ L_{458}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4426. $L_{449} \circ L_{480}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4427. $L_{450} \circ L_{459}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4428. $L_{450} \circ L_{479}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4429. $L_{450} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4430. $L_{450} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4431. $L_{450} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4432. $L_{450} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4433. $L_{451} \circ L_{461}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4434. $L_{451} \circ L_{478}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4435. $L_{451} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4436. $L_{451} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4437. $L_{451} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4438. $L_{451} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4439. $L_{454} \circ L_{462}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4440. $L_{454} \circ L_{475}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (142)(3) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4441. $L_{455} \circ L_{464}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4442. $L_{455} \circ L_{469}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4443. $L_{455} \circ L_{474}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 6 \\ 6 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4444. $L_{455} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4445. $L_{455} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4446. $L_{455} \circ L_{503}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4447. $L_{455} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 5 & 7 \\ 7 & 8 & 7 & 5 \\ 5 & 7 & 8 & 7 \\ 7 & 5 & 7 & 8 \end{pmatrix}$$

4448. $L_{455} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4449. $L_{455} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4450. $L_{455} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4451. $L_{455} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4452. $L_{455} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4453. $L_{455} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 1 & 2 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1432) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4454. $L_{458} \circ L_{480}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (124)(3) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4455. $L_{459} \circ L_{479}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4456. $L_{459} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4457. $L_{459} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4458. $L_{459} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4459. $L_{459} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4460. $L_{461} \circ L_{478}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4461. $L_{461} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4462. $L_{461} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4463. $L_{461} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4464. $L_{461} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (132)(4) \\ \tau_3 = (1234) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4465. $L_{462} \circ L_{475}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (1342) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (123)(4) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \end{pmatrix}$$

4466. $L_{464} \circ L_{503}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 5 \\ 7 & 8 & 5 & 7 \\ 7 & 5 & 8 & 7 \\ 5 & 7 & 7 & 8 \end{pmatrix}$$

4467. $L_{464} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4468. $L_{464} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4469. $L_{464} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4470. $L_{464} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4471. $L_{464} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4472. $L_{464} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4473. $L_{464} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4474. $L_{464} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4475. $L_{464} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1342) \\ \tau_3 = (1243) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4476. $L_{465} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4477. $L_{465} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4478. $L_{465} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4479. $L_{465} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 4 & 3 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 7 & 6 & 8 \\ 7 & 6 & 8 & 7 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4480. $L_{467} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4481. $L_{467} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4482. $L_{467} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4483. $L_{467} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(2)(3) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4484. $L_{469} \circ L_{471}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 5 \\ 7 & 8 & 5 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4485. $L_{469} \circ L_{474}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \\ 6 & 5 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4486. $L_{469} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4487. $L_{469} \circ L_{488}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4488. $L_{469} \circ L_{493}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4489. $L_{469} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4490. $L_{469} \circ L_{495}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4491. $L_{469} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4492. $L_{469} \circ L_{511}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (132)(4) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4493. $L_{469} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4494. $L_{469} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4495. $L_{469} \circ L_{518}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4496. $L_{469} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4497. $L_{469} \circ L_{520}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4498. $L_{469} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4499. $L_{469} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (12)(34) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \\ 5 & 6 & 7 & 8 \end{pmatrix}$$

4500. $L_{469} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4501. $L_{469} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4502. $L_{469} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4503. $L_{469} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4504. $L_{469} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4505. $L_{469} \circ L_{552}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4506. $L_{469} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4507. $L_{469} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4508. $L_{469} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4509. $L_{469} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4510. $L_{469} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 2 & 3 \\ 1 & 2 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4511. $L_{469} \circ L_{569}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4512. $L_{469} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (13)(24) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4513. $L_{471} \circ L_{474}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 5 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4514. $L_{471} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4515. $L_{471} \circ L_{488}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4516. $L_{471} \circ L_{493}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4517. $L_{471} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4518. $L_{471} \circ L_{495}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4519. $L_{471} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4520. $L_{471} \circ L_{511}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4521. $L_{471} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4522. $L_{471} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4523. $L_{471} \circ L_{518}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4524. $L_{471} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4525. $L_{471} \circ L_{520}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4526. $L_{471} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 5 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4527. $L_{471} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4528. $L_{471} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4529. $L_{471} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4530. $L_{471} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4531. $L_{471} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4532. $L_{471} \circ L_{552}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4533. $L_{471} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4534. $L_{471} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4535. $L_{471} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4536. $L_{471} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4537. $L_{471} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4538. $L_{471} \circ L_{569}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \end{pmatrix}$$

4539. $L_{471} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (1324) \\ \tau_3 = (12)(34) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4540. $L_{474} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4541. $L_{474} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4542. $L_{474} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4543. $L_{474} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4544. $L_{474} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (13)(24) \\ \tau_3 = (1234) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4545. $L_{474} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4546. $L_{474} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4547. $L_{474} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4548. $L_{474} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \end{pmatrix}$$

4549. $L_{474} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 5 & 8 \end{pmatrix}$$

4550. $L_{474} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 7 & 6 \\ 6 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4551. $L_{474} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4552. $L_{474} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4553. $L_{478} \circ L_{487}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4554. $L_{478} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4555. $L_{478} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4556. $L_{478} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1234) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 6 & 7 & 7 & 8 \\ 7 & 6 & 8 & 7 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4557. $L_{479} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4558. $L_{479} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4559. $L_{479} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1243) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4560. $L_{479} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (134)(2) \\ \tau_3 = (1243) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4561. $L_{483} \circ L_{491}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4562. $L_{483} \circ L_{562}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4563. $L_{483} \circ L_{571}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (142)(3) \\ \tau_2 = (123)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4564. $L_{487} \circ L_{494}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4565. $L_{487} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 5 \\ 6 & 8 & 6 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4566. $L_{487} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4567. $L_{487} \circ L_{513}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4568. $L_{487} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4569. $L_{487} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 5 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4570. $L_{487} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4571. $L_{487} \circ L_{524}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4572. $L_{487} \circ L_{526}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4573. $L_{487} \circ L_{530}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4574. $L_{487} \circ L_{532}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1432) \\ \tau_2 = (1234) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4575. $L_{487} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4576. $L_{487} \circ L_{540}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4577. $L_{487} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4578. $L_{487} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4579. $L_{487} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 5 & 8 \end{pmatrix}$$

4580. $L_{487} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 5 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 5 & 8 \end{pmatrix}$$

4581. $L_{487} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1432) \\ \tau_2 &= (1234) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4582. $L_{488} \circ L_{493}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4583. $L_{488} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4584. $L_{488} \circ L_{569}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (142)(3) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (142)(3) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4585. $L_{490} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4586. $L_{490} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4587. $L_{490} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4588. $L_{490} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4589. $L_{490} \circ L_{563}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4590. $L_{491} \circ L_{562}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4591. $L_{491} \circ L_{571}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4592. $L_{493} \circ L_{568}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4593. $L_{493} \circ L_{569}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4594. $L_{494} \circ L_{496}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (12)(34) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 5 \\ 6 & 8 & 5 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4595. $L_{494} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \end{pmatrix}$$

4596. $L_{494} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 5 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \end{pmatrix}$$

4597. $L_{494} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4598. $L_{494} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4599. $L_{494} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4600. $L_{494} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4601. $L_{494} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 5 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4602. $L_{494} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4603. $L_{494} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4604. $L_{494} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1423) \\ \tau_2 = (12)(34) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4605. $L_{494} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1423) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4606. $L_{495} \circ L_{566}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4607. $L_{496} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4608. $L_{496} \circ L_{513}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4609. $L_{496} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4610. $L_{496} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 5 & 8 & 6 \\ 5 & 7 & 6 & 8 \end{pmatrix}$$

4611. $L_{496} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4612. $L_{496} \circ L_{524}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (134)(2) \\ \tau_2 &= (123)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4613. $L_{496} \circ L_{526}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(2)(4) \\ \tau_2 &= (1234) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4614. $L_{496} \circ L_{530}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4615. $L_{496} \circ L_{532}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (123)(4) \\ \tau_2 &= (134)(2) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4616. $L_{496} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4617. $L_{496} \circ L_{540}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(2)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(24)(3) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4618. $L_{496} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4619. $L_{496} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4620. $L_{496} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4621. $L_{496} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4622. $L_{496} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4623. $L_{496} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4624. $L_{496} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 2 & 4 & 1 & 3 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(23) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 6 & 8 & 5 & 7 \\ 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \end{pmatrix}$$

4625. $L_{499} \circ L_{501}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4626. $L_{499} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1342) \\ \tau_2 &= (1243) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4627. $L_{499} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4628. $L_{499} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4629. $L_{499} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4630. $L_{499} \circ L_{553}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4631. $L_{499} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4632. $L_{500} \circ L_{502}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1)(234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (143)(2) \\ \tau_2 &= (124)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1)(234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4633. $L_{500} \circ L_{554}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1)(234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (124)(3) \\ \tau_2 &= (143)(2) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1)(234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4634. $L_{500} \circ L_{556}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (14)(2)(3) \\ \tau_2 &= (1243) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1)(234) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (132)(4) \\ \tau_4 &= (1)(234) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4635. $L_{501} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4636. $L_{501} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4637. $L_{501} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4638. $L_{501} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4639. $L_{501} \circ L_{553}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4640. $L_{501} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4641. $L_{502} \circ L_{554}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (132)(4) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4642. $L_{502} \circ L_{556}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (143)(2) \\ \tau_2 = (124)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4643. $L_{503} \circ L_{512}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4644. $L_{503} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4645. $L_{503} \circ L_{522}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4646. $L_{503} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4647. $L_{503} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4648. $L_{503} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4649. $L_{503} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4650. $L_{503} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4651. $L_{503} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 2 & 3 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (14)(23) \\ \tau_2 = (1243) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4652. $L_{506} \circ L_{516}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4653. $L_{506} \circ L_{537}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4654. $L_{506} \circ L_{550}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4655. $L_{511} \circ L_{520}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4656. $L_{511} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4657. $L_{511} \circ L_{552}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (132)(4) \\ \tau_2 = (124)(3) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4658. $L_{512} \circ L_{517}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 5 & 6 & 8 \end{pmatrix}$$

4659. $L_{512} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4660. $L_{512} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4661. $L_{512} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 5 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4662. $L_{512} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4663. $L_{512} \circ L_{553}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4664. $L_{512} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4665. $L_{512} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4666. $L_{512} \circ L_{563}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4667. $L_{512} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4668. $L_{512} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1342) \\ \tau_2 = (1243) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4669. $L_{513} \circ L_{540}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4670. $L_{513} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4671. $L_{513} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 1 & 3 & 4 \\ 1 & 3 & 4 & 2 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4672. $L_{516} \circ L_{537}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4673. $L_{516} \circ L_{550}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4674. $L_{517} \circ L_{519}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(34) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 5 & 7 \\ 6 & 8 & 7 & 5 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4675. $L_{517} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4676. $L_{517} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4677. $L_{517} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4678. $L_{517} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 5 \\ 5 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4679. $L_{517} \circ L_{553}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (12)(34) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4680. $L_{517} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4681. $L_{517} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4682. $L_{517} \circ L_{563}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(2)(3) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(23)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4683. $L_{517} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4684. $L_{517} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 6 & 8 \end{pmatrix}$$

4685. $L_{517} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4686. $L_{518} \circ L_{543}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (13)(24) \\ \tau_2 &= (12)(3)(4) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(3)(4) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(34) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4687. $L_{519} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (1432) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4688. $L_{519} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1324) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4689. $L_{519} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (13)(24) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4690. $L_{519} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (1342) \\ \tau_3 &= (14)(23) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 6 & 8 & 7 & 5 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4691. $L_{519} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1243) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (1342) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4692. $L_{519} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (14)(23) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 6 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4693. $L_{519} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (12)(34) \\ \tau_2 &= (1423) \\ \tau_3 &= (1324) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4694. $L_{519} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 4 & 2 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1324) \\ \tau_2 &= (12)(34) \\ \tau_3 &= (1423) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{aligned} \tau_1 &= (1234) \\ \tau_2 &= (1432) \\ \tau_3 &= (13)(24) \\ \tau_4 &= (1)(2)(3)(4) \end{aligned}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 6 & 8 & 6 \\ 5 & 7 & 6 & 8 \end{pmatrix}$$

4695. $L_{520} \circ L_{541}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4696. $L_{520} \circ L_{552}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1324) \\ \tau_2 = (12)(3)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4697. $L_{522} \circ L_{535}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 6 \\ 6 & 6 & 7 & 8 \end{pmatrix}$$

4698. $L_{522} \circ L_{542}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4699. $L_{522} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4700. $L_{522} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4701. $L_{522} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4702. $L_{522} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 1 \\ 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(24) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4703. $L_{523} \circ L_{525}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4704. $L_{523} \circ L_{529}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4705. $L_{523} \circ L_{531}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4706. $L_{524} \circ L_{526}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4707. $L_{524} \circ L_{530}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4708. $L_{524} \circ L_{532}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4709. $L_{524} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4710. $L_{524} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (134)(2) \\ \tau_2 = (123)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4711. $L_{525} \circ L_{529}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4712. $L_{525} \circ L_{531}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4713. $L_{526} \circ L_{530}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4714. $L_{526} \circ L_{532}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4715. $L_{526} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4716. $L_{526} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 2 & 1 & 3 \\ 3 & 1 & 2 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (13)(2)(4) \\ \tau_2 = (1234) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4717. $L_{529} \circ L_{531}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (142)(3) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4718. $L_{530} \circ L_{532}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4719. $L_{530} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4720. $L_{530} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4721. $L_{532} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4722. $L_{532} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (134)(2) \\ \tau_3 = (1432) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \end{pmatrix}$$

4723. $L_{535} \circ L_{544}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 5 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4724. $L_{535} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 6 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 7 & 8 \end{pmatrix}$$

4725. $L_{535} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4726. $L_{535} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4727. $L_{535} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (13)(24) \\ \tau_3 = (1432) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4728. $L_{537} \circ L_{550}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (143)(2) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 6 & 7 \\ 7 & 6 & 7 & 8 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \end{pmatrix}$$

4729. $L_{540} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 7 & 8 & 6 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4730. $L_{540} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(2)(4) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(24)(3) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 6 & 6 \\ 6 & 7 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4731. $L_{541} \circ L_{552}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 1 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 3 & 2 & 1 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (132)(4) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4732. $L_{542} \circ L_{567}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1324) \\ \tau_3 = (1423) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 7 \\ 5 & 8 & 7 & 7 \\ 7 & 7 & 8 & 5 \\ 7 & 7 & 5 & 8 \end{pmatrix}$$

4733. $L_{544} \circ L_{551}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 6 & 7 \\ 6 & 8 & 7 & 6 \\ 6 & 7 & 8 & 6 \\ 7 & 6 & 5 & 8 \end{pmatrix}$$

4734. $L_{544} \circ L_{553}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4735. $L_{544} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4736. $L_{544} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4737. $L_{544} \circ L_{563}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4738. $L_{544} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4739. $L_{544} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ 2 & 3 & 4 & 1 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (13)(24) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4740. $L_{551} \circ L_{553}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4741. $L_{551} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4742. $L_{551} \circ L_{560}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 8 & 7 & 6 \\ 6 & 7 & 8 & 7 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4743. $L_{551} \circ L_{563}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 7 & 8 & 7 \\ 7 & 8 & 7 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4744. $L_{551} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4745. $L_{551} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 4 & 2 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (1342) \\ \tau_3 = (14)(23) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$

4746. $L_{553} \circ L_{555}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (1342) \\ \tau_4 = (1)(23)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \end{pmatrix}$$

4747. $L_{554} \circ L_{556}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (124)(3) \\ \tau_2 = (143)(2) \\ \tau_3 = (132)(4) \\ \tau_4 = (1)(234) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(2)(3) \\ \tau_3 = (132)(4) \\ \tau_4 = (1)(234) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 7 & 7 & 6 & 8 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \end{pmatrix}$$

4748. $L_{560} \circ L_{565}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 5 & 8 \end{pmatrix}$$

4749. $L_{560} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 3 & 4 & 2 & 1 \\ 1 & 2 & 4 & 3 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1243) \\ \tau_2 = (14)(23) \\ \tau_3 = (1342) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 6 \\ 7 & 8 & 6 & 7 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4750. $L_{562} \circ L_{571}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 3 & 1 & 4 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (134)(2) \\ \tau_4 = (1)(243) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 7 & 7 & 6 \\ 7 & 6 & 8 & 7 \\ 6 & 7 & 7 & 8 \\ 7 & 8 & 6 & 7 \end{pmatrix}$$

4751. $L_{565} \circ L_{570}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 3 & 2 & 4 & 1 \\ 2 & 3 & 1 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(34) \\ \tau_2 = (14)(23) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 2 & 4 \end{pmatrix} \quad \begin{matrix} \tau_1 = (1234) \\ \tau_2 = (1432) \\ \tau_3 = (13)(24) \\ \tau_4 = (1)(2)(3)(4) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 5 & 7 & 6 \\ 6 & 8 & 6 & 7 \\ 7 & 6 & 8 & 5 \\ 6 & 7 & 6 & 8 \end{pmatrix}$$

4752. $L_{568} \circ L_{569}$

$$A = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 1 & 4 & 2 & 3 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (12)(3)(4) \\ \tau_2 = (1423) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$B = \begin{pmatrix} 4 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix} \quad \begin{matrix} \tau_1 = (123)(4) \\ \tau_2 = (142)(3) \\ \tau_3 = (1324) \\ \tau_4 = (1)(2)(34) \end{matrix}$$

$$A \otimes B = \begin{pmatrix} 8 & 6 & 7 & 7 \\ 6 & 8 & 7 & 7 \\ 7 & 7 & 8 & 6 \\ 7 & 7 & 6 & 8 \end{pmatrix}$$